PESO-LINK MISOR: A MOBILE-BASED EMPLOYMENT REGISTRATION AND VALIDATION SUPPORT SYSTEM FOR PESO MISAMIS ORIENTAL

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UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

CAGAYAN DE ORO

JOEREMY CHRISTIE ONG

GON GONOS

VINCE ROY LIM

MILLARD FIL PEJANA

BENE GABRIEL ROSEL

BACHELOR OF SCIENCE IN INFORMATION TECHNOLOGY

COLLEGE OF INFORMATION TECHNOLOGY AND COMPUTING

UNIVERSITY OF SCIENCE AND TECHNOLOGY OF

SOUTHERN PHILIPPINES

MARCH 2026

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CHAPTER I
INTRODUCTION

1.1 Background of the Study

Employment facilitation plays a significant role in connecting job seekers and employers within local communities. As labor markets continue to evolve in response to technological advancements, public employment services increasingly rely on digital platforms to improve accessibility, coordination, and service delivery. While global and national frameworks promote digital transformation in public services, the effectiveness of such initiatives ultimately depends on their implementation at the local government level. In this context, examining how digital tools can enhance employment facilitation within specific provinces becomes essential.

The 2030 Agenda for Sustainable Development emphasizes the need for "full and productive employment and decent work for all," as articulated in Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth. In line with this goal, international labor and development organizations highlight that digital and mobile-enabled employment platforms play a significant role in expanding access to decent work, particularly for youth and underserved populations. As global mobile penetration continues to increase, digital employment facilitation systems provide opportunities to enhance job accessibility, streamline service delivery, and improve communication between job seekers and employers (International Telecommunication Union, 2024).

In the Philippines, the Philippine Statistics Authority (PSA) reported an employment rate of approximately 96.1% and an unemployment rate of 3.9% in September 2025 (Philippine Statistics Authority, 2025). National strategies such as the Trabaho Para sa Bayan (TPB) Plan emphasize strengthening labor-market resilience, improving digital service delivery, and enhancing coordination through technology-enabled employment platforms

(International Labour Organization, 2025). These national directions support the continued modernization of employment facilitation systems at the local government level.

In Region X (Northern Mindanao), employment facilitation services are delivered through Public Employment Service Offices (PESO), including PESO Misamis Oriental. To understand the current provincial workflow, the researchers conducted a semi-structured interview on October 23, 2025 with PESO Manager Mr. Jerome Mundo, assisted by IT Head Ms. Christi Samson. During this interview, PESO personnel described their existing systems for vacancy dissemination, job seeker registration, and employer coordination.

PESO Misamis Oriental maintains an official website and Facebook page for publishing job vacancies and public announcements. Additionally, PESOs nationwide utilize the national PESO Employment Information System (PEIS/PhilJobNet), administered by the Department of Labor and Employment (DOLE), for encoding job seeker and employer information. PESO personnel explained that a web-based platform (Version 1) has been developed for employment facilitation; however, its public accessibility has been intermittent due to cybersecurity-related restrictions. At the time of observation, an enhanced Version 2 of the system was reported to be under development in coordination with the Provincial MIS Office to address these limitations. Despite these developments, no dedicated mobile-native application was available to provide smartphone-optimized registration, vacancy browsing, and application tracking.

Operational data provided by PESO personnel indicate that the office accommodates approximately 1 to 5 walk-in job seekers daily (10 to 32 monthly). Reported weekly vacancy dissemination varies depending on recruitment cycles and employer participation, ranging from approximately 150 available job positions during regular periods to aggregated vacancy counts reaching up to approximately 3,000 available positions during large-scale recruitment activities such as job fairs. The office also coordinates with an estimated 5 to 26 partner employment providers each month. These figures indicate that while walk-in engagement is relatively limited, a high volume of employment opportunities is disseminated digitally through online platforms.

To validate and document actual practices, the researchers conducted an on-site observation and follow-up interview on February 20, 2026 at the PESO Misamis Oriental office. During this visit, the step-by-step workflow of vacancy posting, employer coordination, and applicant engagement was observed and documented. It was noted that job postings are largely disseminated through Facebook announcements. Interested applicants may either send their resumes directly to employers via email or submit them to PESO for referral, depending on employer preference. This dual routing process was confirmed through observation and clarification questions with PESO staff.

To further validate and document actual PESO operational practices, the researchers conducted an on-site observation and follow-up interview on April 20, 2026 at the PESO Misamis Oriental office. The activity provided direct insight into the existing employment facilitation workflow, including job seeker registration, job posting dissemination, application handling, and tracking processes. In the current PESO Misamis Oriental workflow, job seekers provide their information through forms or resumes, after which PESO personnel manually encode and process the data for internal records and coordination with employers. Job vacancies are disseminated primarily through Facebook and similar platforms, while applications are submitted either directly to employers via email or through PESO referral. Tracking of applicant progress is performed manually using communication channels such as email, SMS, or calls, without a centralized system for monitoring application status. As a result, the workflow remains fragmented, with limited visibility of application progress and continued reliance on repetitive manual data handling.

In addition, an interview was conducted on February 24, 2026 with a partner employer of PESO Misamis Oriental to understand the employer-side workflow in job posting and applicant management. The employer reported that documentary requirements such as letter of intent, business permit, and company registration are submitted either physically or online prior to submission and verification of requirements with PESO for job posting. After PESO verifies the submitted requirements, job descriptions are typically sent via email to PESO personnel for posting. Applicant resumes are then received through email and manually sorted by the employer. Monitoring of applicants is conducted manually through email review, and there is no centralized tracking system for document status or application counts. The employer expressed that a more organized and centralized digital platform for application monitoring and job posting management would improve efficiency and reduce manual checking.

In addition to interviews and on-site observation, a survey involving fifty-one (51) respondents from Misamis Oriental was conducted using a five-point Likert scale (1 = Strongly Disagree to 5 = Strongly Agree) to assess user behavior and perceptions regarding employment facilitation processes. Respondents were selected using a non-probability convenience sampling approach within Misamis Oriental due to accessibility and time constraints. The findings are therefore interpreted as indicative rather than statistically generalizable to the entire provincial population.

Results show that thirty-three (33) respondents or 64.7% reported never using PESO services, while ten (10) respondents or 19.6% reported rarely using the services (once or twice a year). Eight (8) respondents or 15.7% reported occasional use. Overall, forty-three (43) out of 51 respondents (84.3%) reported either never using, rarely using, or only occasionally using PESO services. This distribution indicates limited structured access to PESO services among surveyed participants.

When asked about willingness to use a localized PESO-integrated mobile application for job searching, the statement obtained a mean rating of 4.16 out of 5, indicating strong agreement. Forty-two (42) respondents or 82.4% rated the statement as 4 or 5, reflecting positive user response toward mobile-based employment facilitation.

Feature importance ratings further support this readiness. Search and filtering of job postings obtained a mean score of 4.43, application status tracking obtained a mean score of 4.43, and push notifications for job updates received a mean score of 4.39. All mean values exceed 4.0, indicating strong perceived importance of centralized vacancy browsing, structured tracking, and notification-based update mechanisms.

Respondents also agreed that a mobile application would improve PESO's efficiency and community service, with a mean rating of 4.43. Forty-four (44) respondents or 86.3% rated this statement as 4 or 5. These findings indicate positive perception and potential acceptance of a mobile-based employment facilitation approach among surveyed participants.

To further validate these results, brief follow-up interviews were conducted with eighteen (18) selected respondents who consented to additional discussion after completing the survey. Thirteen (13) participants (72.2%) reported uncertainty regarding application status after submission. Twelve (12) participants (66.7%) expressed preference for notification-based updates to reduce manual follow-ups. Several respondents also mentioned that job postings on social media platforms are often scattered and difficult to monitor systematically.

The alignment between quantitative mean ratings and recurring qualitative themes indicates convergence of findings. High importance ratings for application status tracking (mean = 4.43) and push notifications (mean =

4.39) correspond with interview responses emphasizing uncertainty and the need for structured updates.

While the survey did not directly measure encoding difficulty as a standalone variable, follow-up interviews with PESO personnel identified repetitive manual encoding of job seeker information as part of the current registration and data handling workflow. In response, the proposed system adopts an NSRP-based digital profile structure aligned with PESO's job seeker registration process. Optical Character Recognition (OCR) is incorporated only as an optional assistive feature to help extract information from uploaded NSRP form images and pre-fill profile fields, subject to user review, editing, and confirmation. The OCR component will function solely as a data extraction support tool and will not automate screening, ranking, or validation decisions, which remain under PESO personnel authority.

Taken together, the interviews, on-site observation, and survey findings indicate an opportunity to provide a mobile-based extension of existing employment facilitation services at the provincial level. While existing web-based and social media channels provide foundational functionality, there remains no dedicated mobile-based platform that structures registration, vacancy browsing, and application tracking within a single interface.

In recent years, mobile applications have increasingly incorporated document digitization technologies such as Optical Character Recognition (OCR) to reduce manual data entry. OCR technology enables the extraction of text from uploaded document images and converts it into editable digital fields. In this study, OCR is positioned as an optional support mechanism for NSRP-based profile encoding, allowing uploaded NSRP form images to assist in pre-filling selected profile fields. All extracted information remains subject to review and confirmation, ensuring that the feature supports encoding without replacing PESO's official validation and registration authority.

In addition to supporting structured profile encoding through Optical Character Recognition (OCR), employment platforms increasingly incorporate skill-based matching mechanisms to assist job seekers in reviewing how their qualifications relate to job requirements. In this study, skill matching refers to the rule-based comparison of confirmed skills from the job seeker's NSRP-based profile with the required skills encoded in a job posting. The system identifies which required skills are present in the job seeker's profile and which required skills are missing. The output is limited to matched and missing skills and does not include applicant ranking, recommendation, automatic screening, scoring, or hiring decisions.

These systems typically use keyword detection or rule-based similarity comparison to identify overlaps between skill sets. In the proposed system, skill matching is applied only after the user confirms the skills encoded in the NSRP-based profile, ensuring that the skill alignment result is based on reviewed user information rather than raw OCR output.

The system contribution and innovation of this study lie in consolidating PESO Misamis Oriental's fragmented employment facilitation activities into a structured Android-based mobile workflow while preserving PESO and employer decision-making authority. In the existing workflow documented through interviews and observation, job vacancies are mainly disseminated through Facebook or related online channels, applications may proceed through employer email or PESO referral, applicant monitoring is handled manually through email, SMS, calls, or direct coordination, and job seeker information is manually encoded for records and referral processing. In the proposed process, job seekers can create an NSRP-based digital profile, optionally use OCR-assisted form-to-profile encoding, browse localized job postings with search and filtering, view employer-provided application instructions, request PESO referral through the system, and monitor PESO-Link application/referral status updates. Employers can manage job postings and view PESO-referred applicant records within the system workflow, while PESO Admin can manage accounts, monitor postings, review referral requests, facilitate referral routing, and update referral-related status records. The innovation is therefore not automated hiring, applicant ranking, or replacement of PESO procedures, but the integration of localized mobile access, NSRP-based profile encoding, optional OCR assistance, rule-based matched/missing skill indication, and PESO referral tracking in one system-supported workflow aligned with existing PESO procedures.

1.2 Statement of the Problem

1.2.1 General Problem

Employment facilitation in PESO Misamis Oriental is currently characterized by fragmented job information dissemination, absence of structured application tracking, and reliance on manual and repetitive data handling processes across multiple unintegrated channels. These conditions result in limited visibility of applicant progress, inefficient coordination between job seekers, employers, and PESO personnel, and increased workload due to repeated encoding of applicant information across systems.

1.2.2 Specific Problems

a. **Fragmented and Unstructured Job Information Access**

   Job vacancies are primarily disseminated through Facebook and similar platforms, where postings are unstructured, not filterable, and difficult to systematically browse, limiting efficient access to relevant job opportunities.

b. **Lack of Application Tracking and Status Visibility**

   Applicants do not have access to a system that provides confirmation, status updates, or progress tracking after submission, as application monitoring is currently performed manually through email, SMS, or direct communication, resulting in limited visibility of application status.

c. **Manual and Decentralized Application and Coordination Workflow**

Application submission, employer coordination, and applicant processing are conducted through separate channels such as email and direct communication, without a centralized system for managing applications, resulting in fragmented workflow and increased reliance on manual coordination processes.

d. **Repetitive Manual Data Entry During Job Seeker Registration and Data Handling**

Job seeker information, particularly through standardized forms such as NSRP, is manually encoded during registration and processing, often requiring repeated data entry and verification across systems, resulting in increased workload and inefficiency in handling applicant information.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of this study is to design and develop a mobile-based employment facilitation system that provides structured job information access, application tracking, and workflow support for PESO Misamis Oriental.

1.3.2 Specific Objectives

a. To design and develop a mobile-based employment facilitation system that provides structured and localized job vacancy browsing, application tracking, notification-based updates, and centralized application routing for PESO Misamis Oriental.

b. To integrate NSRP-based digital profile encoding with optional OCR-assisted data extraction and rule-based skill matching as support features within the proposed system.
c. To test and evaluate the developed system in terms of functionality and usability using standard evaluation metrics.

1.4 Significance of the Study

This study is significant because it demonstrates a structured mobile-based support model for PESO Misamis Oriental’s employment facilitation workflow. Its contribution is the integration of job seeker registration, localized vacancy browsing, structured PESO referral routing and tracking, employer account management, job posting support, NSRP-based profile encoding, optional OCR-assisted auto-fill, and rule-based matched/missing skill indication within one Android-based system. This contribution differs from the existing PESO workflow because the current process relies on separate channels such as Facebook, email, manual coordination, and manual encoding, while the proposed system organizes these supported functions into a single mobile-accessible workflow without replacing PESO validation authority or employer hiring decisions.

For PESO Misamis Oriental, the proposed system provides a mobile-accessible platform that organizes job seeker registration, vacancy browsing, employer account management, and job posting coordination into a structured workflow. The inclusion of administrative monitoring functionality supports existing manual validation procedures while improving visibility, status tracking, and coordination within a single interface.

For Job Seekers, the system enhances accessibility to employment opportunities by providing localized vacancy browsing, structured search and filtering mechanisms, and simplified application routing options. Given that vacancies are primarily disseminated through social media, consolidating postings within a mobile application improves discoverability, organization, and access to job-related information. Notification alerts further support updates regarding PESO referral status and job announcements. The system also incorporates a rule-based skill matching mechanism that compares the job seeker's confirmed profile skills with the required skills in a job posting. By presenting matched and missing skills, the system allows users to review which required skills are present and which are missing. This feature functions only as a decision-support tool and does not replace employer evaluation or hiring decisions.

For Employers, the proposed system offers a structured method for managing job postings and applicant review within a centralized mobile interface. This supports clearer posting coordination and allows employers to view PESO-referred applicant records when applicable.

For PESO Personnel (Admin Users), the system provides a simplified mobile interface for managing employer accounts, monitoring job postings, and monitoring PESO referral records. By modeling vacancy management digitally, the system supports administrative monitoring and coordination while retaining manual decision-making authority in accordance with existing policies and regulations.

For Local Government Units and Policy Makers, this study demonstrates a feasible approach to extending employment facilitation services through mobile-based technology without replacing existing national systems or requiring full integration with external government databases. The project highlights a practical model for improving localized employment access within a controlled academic environment.

For Future Researchers and Developers, this study establishes a foundation for further exploration of advanced employment facilitation features such as enhanced notification systems, broader system integration, or scalable cloud-based deployment models, while clearly defining the feasibility boundaries of a student-developed mobile system.

Overall, this research supports the continued modernization of public employment services by introducing a structured and accessible mobile application designed to complement and enhance existing PESO workflows.

1.5 Scope of the Study

This study focuses on the design and development of an Android-based mobile-native employment facilitation system for PESO Misamis Oriental. The system supports structured job information access, PESO referral tracking, and centralized workflow management within a single mobile-accessible platform. Core system features include job seeker registration, localized vacancy browsing with search and filtering, employer-provided application instruction viewing, PESO referral request tracking, notification alerts, and employer account management under PESO administrative control, while OCR-based NSRP form-to-profile auto-fill is provided as an optional assistive feature that users may choose to use or skip during profile creation.

For job seekers, the system provides account registration, profile creation, and access to job vacancies with search and filtering functionalities. It allows job seekers to view employer-provided application instructions for external applications and to request PESO referral through the system after completing the required NSRP-based applicant profile fields. Referral tracking and notification alerts apply only to PESO-Link referral records, system-handled application status updates, and job posting updates. Within the system workflow, status tracking may display labels such as Submitted, For Review, PESO-Referred, For Interview, Hired, Rejected, or Closed, depending on the current processing stage and the authorized user who updates the record. The system includes an optional NSRP form-to-profile auto-fill feature using Optical Character Recognition (OCR) to assist users in encoding profile information. The OCR feature is optional and intended only to assist in pre-filling selected NSRP-based profile fields when an accomplished NSRP form image is available; users may still complete the profile manually. The OCR workflow follows these steps: the user uploads an image of the accomplished NSRP form, the system extracts text and pre-fills selected profile fields, and the user reviews, edits, and confirms the extracted information before saving. Manual digital encoding remains available, and all extracted data requires user review, editing, and confirmation before saving. The feature functions solely as an optional assistive encoding tool.

The system supports two application paths. First, job seekers may apply through employer-provided email or application instructions shown in the job posting; this process is handled outside PESO-Link and is not fully tracked by the system. Second, job seekers may request PESO referral after completing the required NSRP-based applicant profile fields. This creates a referral record that can be reviewed by PESO Admin and, when endorsed, shown to the employer with a PESO-Referred label. Final evaluation, document requirements, interviews, and hiring decisions remain under the employer.

The system implements a rule-based skill matching mechanism that compares confirmed NSRP-based profile skills with job posting requirements to provide a basic indication of alignment. This feature functions solely as a decision-support tool and does not perform automated evaluation, recommendation, or automatic screening. Skill matching is available after the user completes or confirms profile skills and is presented as a basic alignment indicator for each job posting. A partial or incomplete skill match does not automatically prevent a job seeker from viewing job details, following employer-provided application instructions, or requesting PESO referral through the system. Referral requests remain subject to PESO Admin review and existing PESO operational procedures.

For employers, the system supports account management and job posting functionalities following existing PESO operational procedures. Employer accounts are created and managed under administrative control, and access is provided upon approval by PESO personnel. Employers do not perform self-registration within the system. Administrative users are able to manage employer accounts, oversee job postings, and monitor PESO referral routing processes.

1.6 Limitations of the Study

The system is limited to the implementation of a mobile-based employment facilitation platform for PESO Misamis Oriental and does not replace existing PESO systems or official decision-making processes. The system does not replace official NSRP processing, PESO validation procedures, or encoding into external government systems such as PEIS or PhilJobNet. The application supports Android devices only and does not include a web-based or iOS version. This limitation is defined to ensure feasibility, focused system development, and complete implementation within the scope and time constraints of the study.

The OCR feature is limited to extracting text from uploaded NSRP form images and may not achieve complete accuracy depending on image quality, formatting, and layout. Extracted information, including detected skills, requires user verification before being saved to the profile.

The skill matching mechanism is rule-based and relies on keyword comparison between job seeker skills and job requirements. It does not use machine learning, artificial intelligence, or predictive analytics, and therefore does not provide automated hiring decisions or recommendations.

The system provides notification alerts related to job postings and PESO referral status updates but does not include real-time messaging, interview scheduling, or direct communication between employers and applicants.

Applications submitted through employer-provided email or external instructions are outside the system's full tracking capability. PESO-Link tracks only referral requests submitted through the system. The system also does not replace employer document requirements, interviews, or hiring decisions.

Evaluation of the system is limited to usability and functionality testing and does not include long-term deployment assessment or labor market impact analysis.

CHAPTER II
REVIEW OF RELATED LITERATURE

This chapter presents the theoretical foundations, empirical studies, and existing systems relevant to the development of the Mobile-Based Employment Registration and Validation Support System for PESO Misamis Oriental. The chapter is organized into five sections: Related Literature (2.1), Related Studies (2.2), Related Systems (2.3), Synthesis (2.4), and System Workflow (2.5). The purpose of this chapter is to establish the academic grounding of the study, identify documented patterns in digital employment facilitation systems, and clearly define the gap addressed by the proposed provincial-level mobile-native system.

2.1 Related Literature

2.1.1 Mobile-First E-Governance

The global transition toward mobile-first governance has significantly reshaped public service delivery. The International Telecommunication Union (2024) reports that mobile connectivity continues to expand worldwide, particularly in developing countries where smartphones serve as the primary access point to digital services.

In the Philippine context, the 2024 National ICT Household Survey (NICTHS) conducted by the Department of Information and Communications Technology in partnership with the Philippine Statistics Authority reported that in Region X (Northern Mindanao), mobile phone usage reached 78.4%, while household internet access reached 54.9%. These figures indicate that mobile-accessible systems are important for ensuring inclusive access to digital government services, especially in regions where smartphones are more prevalent than fixed broadband connections.

In addition, Android remains the dominant mobile operating system globally. A study by Jošt and Taneski (2025) reported that Android accounts for approximately 72.04% of the global smartphone market share as of Q4 2024, compared to iOS at 27.49%, with both platforms collectively accounting for 99.5% of the global market. These findings indicate the numerical dominance of Android over iOS. This dominance is largely attributed to its open-source nature and availability across a wide range of devices, making it more accessible and affordable across different economic segments. In the Philippine context, Android devices also account for a significantly larger share of the smartphone market, reflecting their accessibility among local users (The Report Cube, 2025). Given the high level of mobile usage in Region X, as supported by DICT (2024), Android-based platforms are more suitable for reaching a broader user base, particularly in provincial settings, making them a practical choice for mobile system development within the scope of this study.

The Technology Acceptance Model (TAM), developed by Davis (1989), explains user adoption of information systems through two primary determinants: Perceived Usefulness (PU) and Perceived Ease of Use (PEOU). In employment systems, perceived usefulness may refer to faster job discovery and structured application tracking, while perceived ease of use refers to intuitive navigation and minimized manual encoding.

The Unified Theory of Acceptance and Use of Technology (UTAUT) (Venkatesh et al., 2003) extends TAM by incorporating performance expectancy, effort expectancy, social influence, and facilitating conditions. Jou et al. (2024), in their study of the eGov PH mobile application, found that perceived ease of use and facilitating conditions significantly influenced user adoption in the Philippine setting.

These frameworks support the design of a mobile-based employment system that emphasizes accessibility, usability, and perceived value for local users in Misamis Oriental.

2.1.2 Public Employment Services (PES) Modernization

Public Employment Services globally are evolving from traditional bulletin-board models to digitally supported employment facilitation systems. The International Labour Organization (2025) emphasizes that modern PES must reduce "search frictions," referring to the time and effort required for matching job seekers and employers.

In the Philippines, the PESO Act of 1999 (R.A. 8759) institutionalized Public Employment Service Offices across provinces, cities, and municipalities. More recently, the Trabaho Para sa Bayan (TPB) Plan 2023-2033 (R.A. 11962) highlighted the importance of digital tools in strengthening employment facilitation, especially for Micro, Small, and Medium Enterprises (MSMEs).

National systems such as PhilJobNet provide centralized employment databases administered by the Department of Labor and Employment (DOLE). However, implementation of employment facilitation remains decentralized at the local government level, where operational practices may rely on combinations of websites, social media postings, and manual document submission workflows.

These developments indicate a continuing transition toward digitalization while highlighting the importance of localized system design that complements existing national infrastructure without replacing it.

2.1.3 Optical Character Recognition (OCR) in Mobile Applications

Optical Character Recognition (OCR) technology enables the extraction of text from images and converts it into editable digital format. In employment-related systems, OCR may assist users in transferring structured information from standardized forms, such as NSRP, into digital profile fields more efficiently. This feature is intended only to support the reduction of repetitive manual data entry observed in current registration and processing practices, while still requiring user review and confirmation of all extracted information.

Modern mobile OCR libraries provide on-device text recognition for Android applications. Technical resources indicate that mobile OCR implementations can achieve good accuracy under standard document conditions (Fritz.ai, 2023).

In employment systems, OCR typically functions as an assistive encoding tool rather than a decision-making mechanism. In this study, OCR is incorporated as an optional NSRP form-to-profile auto-fill feature that may extract structured profile information from uploaded NSRP form images, subject to user review and confirmation. It does not perform automated screening, AI-based candidate ranking, or document authenticity verification. Its role is limited to reducing manual data entry during profile creation.

2.1.4 Skill-Based Job Matching in Employment Platforms

Skill-based job matching systems are used in digital recruitment platforms to improve the efficiency of employment search and applicant-job alignment (Modak et al., 2024). These systems compare the skills and qualifications of job seekers with the competency requirements specified in job postings in order to determine how applicant qualifications relate to available job opportunities based on required skills. In local academic employment systems, skill-based filtering has also been used to compare applicant qualifications with employer needs, as demonstrated in JOFS, which filters applicants based on skills, level, and rating (Adaza et al., 2018). Basic implementations commonly use keyword-based or similarity-based comparisons between applicant qualifications and job requirements, while more advanced methods extend these approaches using natural language processing and machine learning techniques (Modak et al., 2024).

In this study, skill matching refers to the rule-based comparison of confirmed job seeker skills and job posting required skills to identify matched and missing required skills. The skill matching mechanism is anchored on the basic keyword-based comparison approach discussed by Modak et al. (2024), where applicant qualifications and job requirements are compared using keyword-based or similarity-based methods. It is also differentiated from the skill-based filtering approach demonstrated in JOFS by Adaza et al. (2018), which filters applicants or jobs based on skill-related criteria. For PESO-Link MisOr, the adopted approach is limited to rule-based keyword comparison: the system compares confirmed NSRP-based skills from the job seeker profile with the required skills specified in job postings and displays only matched and missing required skills. A basic skill alignment result is derived by identifying matched and missing skills relative to the required skills listed in the job posting. For example, if a job posting requires five (5) skills and the applicant possesses three (3) of those skills, the system reflects which required skills are matched and which are missing. This mechanism differs from simple skill-based filtering, which only displays jobs containing specific keywords, as it provides an indication of alignment based on matched and missing skills rather than a binary match result. The system does not perform applicant ranking, automated recommendations, automatic screening, or decision-making functions for employers. No numerical match score or automatic minimum skill-match threshold is computed by the system. The matched and missing skills are displayed only as a decision-support output for job seekers, PESO personnel, and employers within the limits of the system workflow. All evaluation and hiring decisions remain under the authority of PESO personnel and partner employers.

Such mechanisms can assist job seekers in reviewing suitable employment opportunities and allow employment platforms to provide decision-support assistance without automating hiring decisions. In public employment systems, skill-based matching can support users by highlighting matched and missing competencies, thereby guiding applicants in understanding how their qualifications relate to job requirements based on matched and missing skills. Integrating skill-based matching with NSRP-based profile encoding and optional OCR-assisted data extraction can further enhance employment systems by reducing manual data entry while providing structured insights into applicant qualifications.

2.1.5 Software Quality and Usability Evaluation

The System Usability Scale (SUS) developed by Brooke (1996) is a widely used tool for evaluating perceived usability. A SUS score above 68 indicates above-average usability, while scores above 80 indicate high usability.

ISO/IEC 25010 provides a standardized framework for evaluating software product quality across characteristics such as Functional Suitability, Reliability, Usability, Security, and Performance Efficiency. Studies applying ISO 25010 in government and mobile systems demonstrate its effectiveness for systematic evaluation (Moumane et al., 2024).

These frameworks guide the evaluation of the proposed mobile application to ensure that usability, reliability, and functional adequacy are measured using recognized standards.

2.2 Related Studies

This section presents empirical studies relevant to mobile employment systems and digital public service adoption.

2.2.1 Web-Based and Mobile Employment Information Systems

Parinsi et al. (2020) developed a job seeker information system using a web-based platform integrated with Android mobile access for the Minahasa Regency Manpower Office. The system was designed using a prototyping development model and allowed job seekers to register, view job vacancies, and submit applications online. The study demonstrated that integrating web and mobile technologies can support access to employment information and online registration and recruitment processes. However, the system primarily focused on online registration and vacancy dissemination and did not emphasize structured employer-side validation mechanisms or mobile-native optimization for localized employment facilitation.

Similarly, Geneta et al. (2019) developed a web-based PESO Job Portal for the Municipality of Rosario, Batangas. The system aimed to support job matching and provide an online alternative to the traditional manual application process. It was evaluated using the ISO 9126 Software Quality Model and received an overall rating of "Excellent" in terms of usability, reliability, and efficiency. While the system enhanced job posting and application processes within a municipal setting, it remained primarily web-based and did not incorporate mobile-native features, structured employer validation workflows, or document-processing enhancements.

These studies demonstrate that web-based employment systems can support service delivery and employment-related processes. However, most documented systems focus on general job posting and applicant registration without explicitly addressing provincial-level mobile-native optimization and structured employer-side validation workflows.

2.2.2 Public Employment Service Awareness and Service Delivery Studies

Gupit, Bajao, and Torrion (2025) assessed the level of awareness among recent graduates regarding the Public Employment Services offered by the Local Government Unit of Tangub City. The study identified varying levels of awareness across service categories such as referral and placement, employment coaching, job fair programs, and labor market information. Notably, labor market information services recorded the lowest awareness among respondents. The findings highlight that despite the presence of public employment services, limited awareness may affect service utilization and employment outcomes.

Awareness studies such as this provide insight into gaps in information dissemination and public engagement. However, they primarily evaluate perception and awareness rather than proposing technological interventions to improve structured service delivery or enhance digital employment facilitation at the provincial level.

2.2.3 Digital Transformation in Public Employment Services

The transition from manual to digital employment systems has been recognized as a key strategy in improving labor market access and administrative efficiency. Digital employment portals provide centralized platforms for job postings, applicant management, and service coordination. Digital employment portals commonly include features such as online registration, job vacancy browsing, and applicant tracking, while some systems may also incorporate skill matching or compatibility indication features (Parinsi et al., 2020; Geneta et al., 2019).

Despite advancements in digital platforms, many existing systems remain web-centric and are not fully optimized for mobile-native user experience. Moreover, documentation on structured employer validation support mechanisms, particularly within provincial-level public employment settings, remains limited. Most systems emphasize job listing and applicant submission processes rather than structured employer-side screening workflows integrated into a mobile-optimized framework (Parinsi et al., 2020; Geneta et al., 2019).

2.2.4 Emerging Support Technologies in Employment Systems

Recent developments in digital systems have explored the integration of assistive technologies such as automated document processing and data extraction tools to improve efficiency in handling applicant documents. Optical Character Recognition (OCR) technology, for instance, has been applied in various information systems to digitize and extract structured data from uploaded documents (Fritz.ai, 2023; Modak et al., 2024). In employment contexts, such tools may assist in streamlining document encoding and validation processes.

However, OCR is generally implemented as a support mechanism rather than the core function of employment systems. Most employment facilitation systems remain focused on vacancy dissemination and applicant registration rather than structured document-processing enhancement integrated within a localized public employment framework (Fritz.ai, 2023; Modak et al., 2024).

2.3 Related Systems

This section focuses primarily on Philippine and local employment systems.

2.3.1 Government Employment Systems

PhilJobNet, administered by DOLE, functions as a national employment database that supports employer registration, job listings, and job-applicant matching. However, it is primarily web-based and not optimized for localized mobile-native engagement or matched/missing skill gap indication as defined in this study.

Local PESO websites typically serve as information portals. Vacancy dissemination frequently relies on Facebook pages and email coordination, which may lack structured filtering and tracking mechanisms.

2.3.2 Local and Regional Capstone Systems

JOFS (Adaza et al., 2018) is a recruitment platform designed to support both job seekers and recruiters through filtering and web crawling technology. The system filters applicants based on factors such as skills, level, and rating, and includes a background-checking feature through web crawling. For job seekers, the system provides job listings aligned with their qualifications, experience, and preferences, while for recruiters, it allows filtering of applicants based on relevant criteria. However, the system is implemented as a web-based platform and does not provide mobile-native functionality or structured employment facilitation within a localized public service context.

PWDJobMatch (Bacasan et al., 2022) is a specialized recruitment system developed to address the marginalization of Persons with Disabilities (PWD) in standard online job listings. It provides an inclusive environment where PWD job seekers can browse job listings and interact with employers. The system supports multiple user roles and basic job search and communication functionalities. However, it does not incorporate OCR-based data extraction or mobile-native system architecture.

Job Alert (Baluran et al., 2018) is an Online Job Fair Management System designed to digitalize traditional job fair processes. The system enables employers to post job vacancies and allows applicants to receive notifications through SMS during recruitment events, as well as participate in online interactions related to job fair activities. However, it is primarily event-based and does not support continuous employment facilitation or structured application tracking outside of job fair activities.

PESO GMA (Marfil et al., 2023) is a system developed to support employment facilitation within a municipal PESO setting. The study implemented a mobile application for applicants and a separate system for administrators and staff, enabling job posting, resume submission, and applicant management. While the system improves accessibility and recruitment processes, it does not incorporate OCR-assisted profile encoding or rule-based skill matching mechanisms.

These systems demonstrate progressive digitalization through mobile applications, web-based recruitment portals, job posting support, filtering, search, and notification functions. However, the reviewed systems do not simultaneously provide localized mobile-native employment facilitation, optional NSRP form-to-profile OCR assistance, and rule-based matched/missing skill gap indication within the specific PESO Misamis Oriental context.

Table 1. Comparison of Existing Systems

Feature	Proposed PESO-Link MisOr	PhilJobNet	PESO GMA (2023)	JOFS	PWDJobMatch
Mobile-Native	✔	✖	✔	✖	✖
Localized Focus	✔	✖	✔	✔	✔
Employer Account Management / Job Posting Support	✔	✔	✔	✔	✔
OCR Profile Assistance	✔	✖	✖	✖	✖
Matched/ Missing Skill Gap Indication	✔	✖	✖	✖	✖
Application/ Referral Handling or Status Visibility	✔	✔	✔	✔	✔
Notification Alerts	✔	✖	✖	✖	✖

In this table, application/referral handling for PESO-Link MisOr refers only to PESO referral tracking within the system; employer-email applications remain external. Matched/missing skill gap indication refers to the rule-based display of matched and missing required skills without scoring, ranking, recommendation, or automated hiring decisions. PhilJobNet provides job-applicant matching, while JOFS and PWDJobMatch support filtering or search functions; however, matched/missing skill gap output was not identified in the reviewed systems.

The comparative analysis highlights the functional differences among existing employment facilitation systems and the proposed PESO Misamis Oriental mobile application. Existing platforms such as PhilJobNet primarily operate as centralized national employment databases that support job posting, applicant registration, and job-applicant matching, but remain largely web-based and do not provide localized mobile-native engagement or matched/missing skill gap indication as defined in this study. Similarly, previously developed academic systems such as JOFS and PWDJobMatch demonstrate the use of web-based recruitment portals with filtering or skill-based search functionalities; however, these systems generally rely on browsing or keyword filtering rather than structured mechanisms that identify matched and missing skills between job seeker qualifications and employer-defined job requirements. Other systems, such as Job Alert, focus on event-based recruitment coordination, including job fair management and notification mechanisms, rather than continuous employment facilitation and applicant–job compatibility assessment.

In contrast, the proposed system integrates several functionalities within a single mobile-native platform designed specifically for the provincial employment context of PESO Misamis Oriental. In addition to localized vacancy browsing and employer account management with administrative control, the system incorporates NSRP-based profile encoding with optional OCR-assisted data extraction and a rule-based skill matching mechanism that compares job seeker skills with employer-defined job requirements. This comparison displays matched and missing required skills based on the job seeker's confirmed profile skills and the job posting requirements. By combining mobile accessibility, employer account management and job posting support under administrative monitoring, NSRP-based profile encoding with optional OCR-assisted data extraction, and rule-based matched/missing skill indication within a unified interface, the proposed system addresses functional gaps identified in existing employment facilitation platforms while remaining within the practical scope of a student-developed mobile application.

In the proposed system, application tracking refers specifically to PESO referral records submitted through PESO-Link. Applications submitted through employer-provided email or external instructions remain outside the system’s full tracking capability.

2.4 Synthesis

The reviewed literature and related systems indicate a continuing transition toward digital and mobile-supported employment facilitation services. Global and national studies highlight the increasing role of mobile-first technologies in improving accessibility to government services, particularly in regions where smartphones serve as the primary access point to the internet. Theoretical frameworks such as the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) further emphasize that perceived usefulness, ease of use, and facilitating conditions significantly influence user adoption of digital platforms, including employment facilitation systems.

Existing employment systems demonstrate various approaches to digital recruitment support. National platforms such as PhilJobNet function primarily as centralized employment databases that allow employers to publish vacancies and job seekers to register profiles. However, these systems generally operate through web-based interfaces and are designed for nationwide data aggregation rather than localized employment facilitation at the provincial level. Similarly, previously developed academic and local systems, including JOFS and PWDJobMatch, illustrate the potential of digital job portals in improving job discovery and applicant registration processes. Despite these advancements, many systems remain web-centric and typically rely on browsing, keyword filtering, or basic search functionalities rather than structured mechanisms that identify matched and missing skills between job seeker qualifications and employer-defined job requirements.

The literature also highlights the increasing adoption of document digitization technologies, such as Optical Character Recognition (OCR), in mobile applications to reduce manual data entry and improve user experience. OCR enables systems to extract textual information from uploaded document images, such as NSRP form images, and convert them into editable digital fields. In employment-related applications, this capability can assist job seekers in transferring structured form information into system profiles more efficiently, thereby helping reduce encoding effort during profile creation.

Despite these developments, the review of existing systems reveals that most employment facilitation platforms do not simultaneously integrate mobile-native accessibility, localized employment focus, employer job posting management, NSRP-based profile encoding with optional OCR-assisted data extraction, and rule-based matched/missing skill indication within a single unified interface. Many platforms emphasize vacancy dissemination and applicant registration but provide limited assistance to job seekers in understanding how closely their qualifications align with available job opportunities.

In response to these identified gaps, the proposed study develops an Android-based mobile employment facilitation application tailored for PESO Misamis Oriental. The system consolidates job seeker registration, localized vacancy browsing, structured PESO referral routing, employer account management, and job posting functionalities within a mobile-native environment. In addition, the system incorporates NSRP-based profile encoding with optional OCR-assisted data extraction, allowing job seekers to upload NSRP form images from which structured profile information may be extracted for user confirmation and editing. To further support job seekers in reviewing job requirements, the system also implements a rule-based skill matching mechanism that compares confirmed job seeker skills with employer-defined job requirements and displays matched and missing required skills. This matching approach is based on the rule-based keyword comparison concept identified in the reviewed skill matching literature and is adapted only to display matched and missing required skills rather than to compute scores, rank applicants, or automate hiring decisions.

Unlike advanced recruitment systems that rely on artificial intelligence for automated hiring decisions, the proposed system functions strictly as a decision-support tool. Employer validation and hiring decisions remain under the authority of PESO personnel and partner employers. By integrating mobile accessibility, NSRP-based profile encoding with optional OCR-assisted data extraction, and rule-based skill matching within a localized employment platform, the proposed system addresses the functional limitations identified in existing employment facilitation systems while remaining within the feasible scope of a student-developed mobile application deployed in a controlled academic environment.

2.5 System Workflow

The PESO-Link MisOr system follows a structured workflow that supports job seekers, PESO administrators, and employers within a centralized mobile-based platform. The workflow reflects the intended operation of the system in improving access to job information, PESO referral tracking, and coordination processes.

For job seekers, the process begins with account registration, followed by profile creation. Users may manually input their information or optionally use the OCR-based NSRP form-to-profile auto-fill feature to assist in encoding profile data. All extracted information is subject to user review, editing, and confirmation before being saved. Once the profile is completed, users can browse available job vacancies using search and filtering functionalities. The system provides a basic skill matching indication by comparing confirmed NSRP-based profile skills with job requirements. Users may either follow employer-provided email or application instructions shown in the job posting, or request PESO referral through the system after completing the required NSRP-based applicant profile fields. Email-based applications proceed outside PESO-Link and are not fully tracked by the system. In PESO-Link, routing refers to the movement of a PESO referral request from the job seeker to PESO Admin for review and, when endorsed, to the employer for viewing within the system workflow. Tracking refers to the monitoring of the recorded application/referral status after submission, including updates shown to the job seeker through status records and notification alerts.

For PESO administrators, the workflow includes the management of employer accounts and job postings. Employer accounts are created and approved by PESO personnel in accordance with existing procedures. Approved employers can publish job postings directly within the system, while PESO Admin monitors postings and may remove inappropriate or invalid postings when necessary. PESO administrators are also able to monitor PESO referral requests, review submitted application/referral records, facilitate referral routing to employers when appropriate, and update referral-related status records within the limits of the system workflow. When PESO Admin updates a referral record, such as marking it as For Review, PESO-Referred, Rejected, or Closed, the updated status is stored in the system and becomes visible to the job seeker through referral status tracking. The system also generates an in-app notification alert to inform the job seeker that the referral record has been processed or updated. These status updates and notification alerts allow applicant progress to be monitored without replacing PESO validation procedures or employer hiring decisions.

For employers, access to the system is provided through administrator-approved accounts. Employers may create and manage job postings, provide application email or instructions, and view applicant information associated with PESO referral records submitted through the platform. The system supports the viewing of applicant profiles and submitted data; however, hiring decisions and further communication processes remain outside the system and follow existing PESO and employer procedures.

This workflow demonstrates the operational difference between the existing PESO workflow and the proposed system-supported workflow. In the existing workflow, job posting dissemination, applicant submission, referral coordination, applicant monitoring, and profile encoding are handled through separate channels such as Facebook, email, SMS, calls, forms, and manual coordination. In the proposed workflow, PESO-Link MisOr organizes these activities into role-based system functions: job seekers create NSRP-based profiles, browse structured job postings, view application instructions, request PESO referral, and track application/referral status; PESO Admin monitors accounts and job postings, reviews referral requests, facilitates routing, and updates referral-related statuses; and employers manage job postings and view PESO-referred applicant records within the limits of the system workflow. The proposed workflow remains system-supported rather than fully automated because PESO validation, referral judgment, employer evaluation, interviews, and hiring decisions remain under authorized human users.

CHAPTER III

METHODOLOGY

This chapter describes the development methodology and process used in building the PESO-Link MisOr system. It also explains how the researchers gathered the necessary data and information used in the design and implementation of the system. This chapter includes the research design and development methodology, requirements gathering and analysis, research locale, participants, system design, software and hardware requirements, research instruments, testing and evaluation procedure, ethical considerations, and data analysis.

3.1 Research Design and Development Methodology

This study uses a developmental research design focused on the design, development, and evaluation of a mobile-based employment facilitation system for PESO Misamis Oriental. The study is developmental since it aims to produce a functional Android-based mobile application that addresses validated workflow issues in employment facilitation, particularly structured job vacancy access, PESO referral request tracking, centralized referral routing support, and NSRP-based profile encoding support.

The development of the system follows an Agile development approach. This approach was selected since the proposed system requires iterative analysis, design, development, testing, and refinement based on validated PESO workflow requirements. The Agile approach allows the researchers to develop the system in manageable stages, while ensuring that each core feature remains aligned with the identified operational problems.

The development process is divided into the following phases:

3.1.1 Requirements Analysis

In this phase, the researchers analyzed the current employment facilitation workflow of PESO Misamis Oriental based on the interviews, on-site observation, survey responses, and follow-up interviews. The analysis focused on identifying the actual workflow used in job posting, job seeker registration, application handling, employer coordination, and application status monitoring.

The requirement analysis was limited to the validated problems identified in the study: fragmented job posting dissemination, absence of structured application tracking, manual and decentralized coordination, and repetitive manual encoding of job seeker information through NSRP-related data handling.

3.1.2 System Design

In this phase, the researchers prepared the system structure, workflow, user roles, database design, and interface flow of the proposed mobile-based employment facilitation system. The system design includes three primary user roles: Job Seeker, Employer, and PESO Admin.

The system design focused on supporting structured job vacancy browsing, job seeker registration, NSRP-based profile creation, optional OCR-assisted NSRP form-to-profile auto-fill, employer-provided application instruction viewing, PESO referral request submission and routing, referral tracking, notification alerts, employer account management, job posting management, and administrative monitoring.

The optional OCR feature was designed only as an assistive tool for extracting text from uploaded NSRP form images and pre-filling profile fields subject to user review and confirmation. It does not perform automated validation, screening, ranking, or hiring decisions. The rule-based skill matching mechanism was also designed only as a decision-support feature that compares confirmed profile skills with job posting requirements.

Based on the rule-based keyword comparison approach anchored in the reviewed skill matching literature, the workflow and logic of the skill matching feature begin after the job seeker has completed or confirmed the skills stored in the NSRP-based profile. These confirmed skills may come from manual encoding or from OCR-assisted extraction; however, OCR-extracted skills are not used for matching unless they are reviewed, edited if necessary, and confirmed by the user. The actual matching process follows a rule-based comparison procedure. First, the system retrieves the confirmed skills stored in the job seeker's NSRP-based profile. Second, the system retrieves the required skills encoded in the selected job posting. Third, each required skill from the job posting is compared against the confirmed job seeker skills. Fourth, if a required skill is present in the confirmed job seeker skills, the system classifies it as a matched skill. Fifth, if a required skill is not present in the confirmed job seeker skills, the system classifies it as a missing required skill. Sixth, the system displays the matched and missing required skills to the user. The process uses only confirmed profile skills and employer-defined job requirements. The system does not compute a numerical match score, does not enforce an automatic minimum skill-match threshold, does not rank applicants, does not recommend applicants, and does not make referral or hiring decisions.

For example, if a job posting requires Communication Skills, Computer Literacy, Customer Service, Microsoft Office, and Data Entry, while the job seeker's confirmed profile skills include Communication Skills, Microsoft Office, and Data Entry, the system displays Communication Skills, Microsoft Office, and Data Entry as matched skills and Computer Literacy and Customer Service as missing required skills. This example illustrates that the system only compares listed skills and displays the result. It does not calculate a score, determine eligibility, approve referrals, or make hiring decisions.

The integration of OCR, digital profile encoding, skill matching, and referral generation follows a sequential system workflow. First, the job seeker creates an NSRP-based digital profile by manually encoding information or by using optional OCR-assisted NSRP form-to-profile auto-fill. If OCR is used, the extracted information is treated only as temporary pre-filled data until the job seeker reviews, edits if necessary, and confirms it. Second, the confirmed profile information, including confirmed skills, is saved as the job seeker's NSRP-based digital profile. Third, when the job seeker views a job posting, the system compares the confirmed profile skills with the required skills encoded in that job posting and displays the matched and missing required skills. Fourth, the job seeker may still view the employer-provided application instructions or request PESO referral through the system. Fifth, when a PESO referral request is submitted, the system generates a PESO-Link application/referral record connected to the job seeker profile, selected job posting, and referral status tracking workflow. This integration allows the system to connect profile encoding, optional OCR assistance, rule-based skill matching, and PESO referral tracking without automatically screening applicants, approving referrals, or making hiring decisions.

3.1.3 System Development

In this phase, the researchers will develop the Android-based mobile application based on the approved system design. The development will focus on implementing the functional modules required by the identified workflow, including account management, job vacancy browsing, employer-provided application instruction viewing, PESO referral routing, referral status tracking, notifications, NSRP-based profile encoding, optional OCR-assisted auto-fill, and administrative monitoring.

The system will be developed as an Android-focused mobile application using React Native with Expo for the frontend, supported by a Node.js Express backend and a MySQL database. Although the Expo development environment may allow developer preview through other platforms, the submitted capstone implementation and testing scope remain limited to Android. The implementation will not include iOS deployment, web-based deployment, real-time messaging, interview scheduling, payment processing, automated hiring decisions, artificial intelligence-based screening, or integration with external government databases.

3.1.4 System Testing

In this phase, the researchers will test whether the developed system performs its intended functions according to the defined requirements. Testing will include checking user registration, login, profile creation, job posting, job browsing, employer-provided application instruction viewing, PESO referral request submission, referral status tracking, notification alerts, employer account management, admin monitoring, optional OCR-assisted encoding, and rule-based skill matching.

The purpose of testing is to verify whether the system functions are working as intended. The testing process will not measure employment success, hiring outcomes, labor market impact, or long-term deployment effectiveness.

3.1.5 System Evaluation

In this phase, the developed system will be evaluated in terms of functionality and usability using appropriate evaluation instruments. Functionality evaluation will determine whether the system features operate according to the stated requirements. Usability evaluation will determine how users assess the system after performing assigned tasks based on their role.

The evaluation will be limited to system-level assessment. It will not claim to prove improved employment outcomes, faster hiring, increased job placement, or improved decision-making unless such outcomes are directly measured.

3.2 Requirements Gathering and Analysis

The requirements of the proposed system were based on the data gathered from PESO personnel, a partner employer, on-site observations, survey respondents, and follow-up interviews. These sources were used to ensure that the proposed system addresses actual workflow issues rather than assumed problems.

The researchers conducted interviews with PESO Misamis Oriental personnel to understand the existing employment facilitation process, including job posting dissemination, job seeker registration, employer coordination, and application handling. The interviews confirmed that PESO uses existing channels such as Facebook, email, and manual coordination to facilitate employment-related transactions.

On-site observations were also conducted to document the actual workflow used by PESO Misamis Oriental. These observations confirmed that job vacancies are commonly disseminated through Facebook and similar platforms, while applications may be submitted directly to employers through email or routed through PESO, depending on the employer's preference. Applicant and referral progress is monitored manually through email, SMS, calls, or direct coordination. Job seekers may wait for updates through manual communication, and there is no structured system-based tracking where they can view recorded application or referral status updates after submission.

A partner employer interview was conducted to understand the employer-side workflow. The employer described the process of submitting job posting details and receiving applicant resumes through email. Applicant review and monitoring are handled manually, and there is no centralized system for tracking applicant status or application counts.

A survey was conducted among fifty-one (51) respondents from Misamis Oriental to assess user behavior and perception regarding employment facilitation. The survey results showed the perceived importance of structured job vacancy browsing, application status tracking, and notification-based updates. Follow-up interviews with selected respondents further supported the need for clearer application status visibility and structured updates after submission.

Based on these findings, the current PESO Misamis Oriental workflow and the proposed PESO-Link MisOr workflow differ in terms of organization, visibility, and system-supported processing. In the current workflow, job postings are disseminated mainly through Facebook and related online channels, job seekers submit applications either through employer-provided email or PESO referral, applicant monitoring is handled manually through email, SMS, calls, or direct coordination, and job seeker information is manually encoded for records and referral processing. In the proposed workflow, these activities are organized into role-based system functions. Job seekers create NSRP-based digital profiles, browse structured job postings, view employer-provided application instructions, request PESO referral through the system, and track PESO-Link application/referral status updates. PESO Admin manages accounts, monitors job postings, reviews referral requests, facilitates referral routing, and updates referral-related status records. Employers manage job postings and view PESO-referred applicant records within the limits of the system workflow. This proposed process remains system-supported rather than fully automated because PESO validation, referral judgment, employer evaluation, interviews, and hiring decisions remain under authorized human users.

Table 2. Comparison of the Current Manual Process and Proposed PESO-Link MisOr Process

Workflow Area	**Current Manual PESO Workflow**	**Proposed PESO-Link MisOr System-Supported Workflow**
Job Posting Dissemination	Job vacancies are commonly disseminated through Facebook and related online channels.	Job postings are organized within the mobile application, allowing job seekers to browse localized vacancies using

	Job seekers need to manually check posts and updates from separate sources.	structured search and filtering functions.
Application Path	Job seekers may submit applications directly through employer-provided email or through PESO referral, depending on employer preference.	Job seekers may still follow employer-provided email or application instructions shown in the job posting for external applications. However, PESO referral requests submitted through PESO-Link are recorded within the system and may be reviewed, routed, and tracked through status updates.
Applicant Monitoring	Applicant and referral progress is monitored manually through email, SMS, calls, or direct coordination. Job seekers may wait for updates through manual communication, and there is no structured system-based tracking where they can view recorded application or referral status updates after submission.	PESO-Link stores referral/application records submitted through the system and allows authorized users to update status labels such as Submitted, For Review, PESO-Referred, For Interview, Hired, Rejected, or Closed.

Referral Routing	PESO referral coordination is handled manually through direct communication between PESO personnel, job seekers, and employers.	PESO Admin can review referral requests, facilitate routing to employers when appropriate, and update referral-related status records within the system workflow.
Job Seeker Information Encoding	Job seeker information is manually encoded from forms or resumes for records and referral processing.	Job seekers create NSRP-based digital profiles. Optional OCR-assisted NSRP form-to-profile auto-fill may assist encoding, but all extracted data must still be reviewed, edited, and confirmed by the user before saving.
Skill Matching	Manual review of applicant qualifications depends on PESO personnel or employer checking.	The system displays rule-based matched and missing required skills based on confirmed job seeker skills and job posting requirements. It does not compute a score, rank applicants, screen applicants, or make hiring decisions.
Decision-Making Authority	PESO personnel and employers manually review referrals, requirements, interviews, and hiring decisions.	PESO personnel and employers still retain validation, referral judgment, interview, and hiring authority. The system only supports organization, visibility, routing, tracking, and record handling.

Based on the gathered data, the following system requirements were identified:

1. The system shall provide structured and localized job vacancy browsing with search and filtering functions to address fragmented job posting access.
2. The system shall provide PESO referral request submission and tracking features to allow job seekers to submit referral requests through PESO-Link and view the recorded status of those referral records after submission.
3. The system shall provide centralized PESO referral routing and administrative monitoring to support the movement of referral records from job seeker submission to PESO Admin review and, when endorsed, to employer viewing within a single workflow.
4. The system shall support NSRP-based digital profile encoding to align with PESO's job seeker registration process.
5. The system shall include optional OCR-assisted NSRP form-to-profile auto-fill to assist users in encoding profile information, subject to user review and confirmation.
6. The system shall provide notification alerts related to job posting updates and PESO referral status updates.
7. The system shall provide employer account management under PESO administrative control.
8. The system shall provide job posting management under PESO administrative monitoring.
9. The system shall allow PESO Admin to monitor job seeker accounts and deactivate accounts with incomplete, invalid, or inappropriate information when necessary.
10. The system shall include rule-based skill matching as a decision-support feature that compares confirmed job seeker skills with job posting required skills and displays the matched

and missing required skills without computing a score, ranking applicants, or making referral or hiring decisions.

The system requirements were derived only from the validated problems and workflow needs identified through the study's data sources. Features that are not supported by the gathered data, such as automated hiring decisions, artificial intelligence-based applicant screening, real-time messaging, payment processing, or interview scheduling, are excluded from the system scope.

3.3 Research Locale and Participants of the Study

3.3.1 Research Locale

The study was conducted in relation to the employment facilitation workflow of PESO Misamis Oriental. The requirement gathering activities were based on interviews and on-site observations involving PESO Misamis Oriental personnel, a partner employer, and survey respondents from Misamis Oriental. The locale was selected because the proposed system is specifically intended to support the employment facilitation process of PESO Misamis Oriental.

3.3.2 Participants of the Study

The participants of the study include PESO personnel, a partner employer, job seekers, and selected survey respondents from Misamis Oriental. These participants were included because they are directly connected to the employment facilitation workflow being studied.

PESO personnel served as key informants for understanding the current workflow of job posting dissemination, job seeker registration, application routing, employer coordination, and application monitoring. Their input provided the operational basis for defining the system requirements and administrative functions of the proposed system.

The partner employer provided information regarding the employer-side process of job posting submission, applicant receiving, resume review, and manual application monitoring. This input supported the design of employer-side job posting and applicant review functions within the proposed system.

The survey respondents consisted of fifty-one (51) individuals from Misamis Oriental. Their responses were used to assess behavior and perception regarding PESO service access, mobile-based employment facilitation, application tracking, vacancy browsing, and notification-based updates. Since the respondents were selected through convenience sampling, the survey findings are treated as indicative and are not presented as statistically generalizable to the entire population of Misamis Oriental.

Selected respondents also participated in follow-up interviews to provide additional explanation regarding their experiences and concerns after submitting job applications. Their responses helped support the need for status visibility and notification-based updates after application or referral submission.

The participants were grouped according to their relevance to the system workflow: PESO personnel for administrative and coordination processes, employers for job posting and applicant review processes, and job seekers or respondents for registration, vacancy browsing, external application instruction viewing, PESO referral requests, and referral tracking processes.

3.4 System Design

This section presents the system design of PESO-Link MisOr, including the system architecture, use case diagram, data flow diagram, entity-relationship diagram, and system workflow. The system design was based on the validated workflow of PESO Misamis Oriental and the identified system requirements. The design focuses on three primary user roles: Job Seeker, Employer, and PESO Admin.

The system design is limited to the features defined in the scope of the study. These include job seeker registration, NSRP-based profile creation, optional OCR-assisted NSRP form-to-profile auto-fill, localized vacancy browsing, search and filtering, employer-provided application instruction viewing, PESO referral request submission and tracking, notification alerts, employer account management, job posting management, and PESO administrative monitoring. The system does not include automated hiring decisions, artificial intelligence-based screening, real-time messaging, payment processing, interview scheduling, or integration with external government databases.

3.4.1 System Architecture

Figure 1 presents the system architecture of PESO-Link MisOr. The architecture shows how the three user roles interact with the Android mobile application and its backend components. The Job Seeker, Employer, and PESO Admin access the system through the mobile application, while the backend processing layer handles authentication, profile management, job posting management, PESO referral tracking, notification processing, optional OCR-assisted profile encoding, rule-based skill matching, database storage, and file storage.

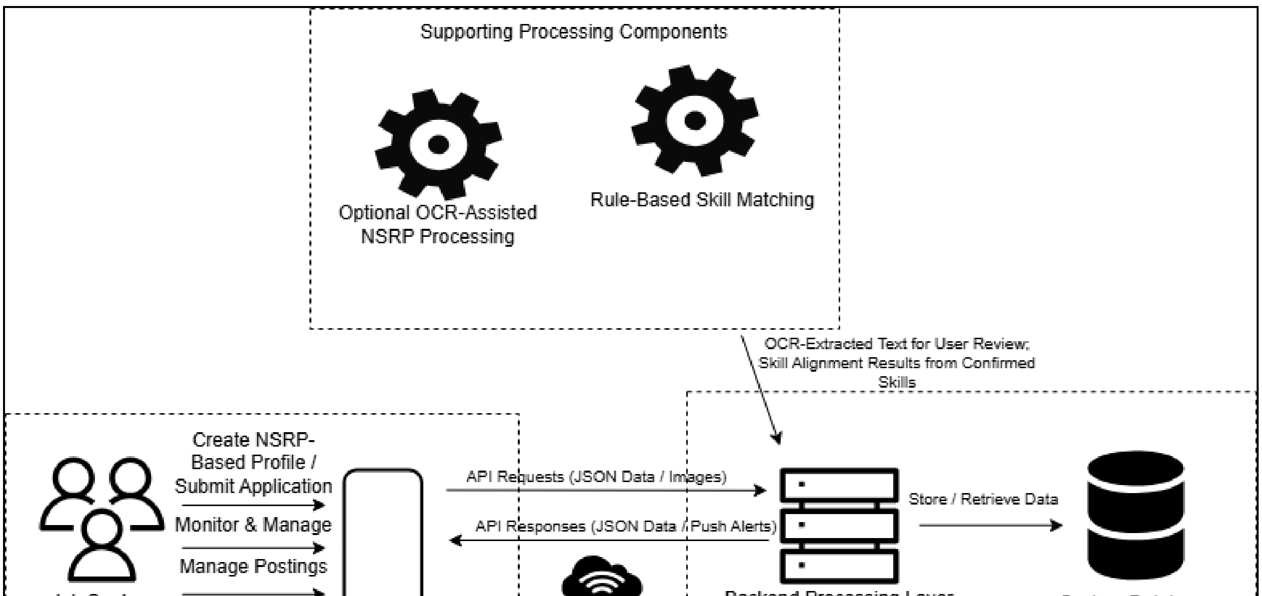

Figure 1. System Architecture of PESO-Link MisOr

Technically, the architecture follows a frontend-backend-database structure. The Android-focused mobile application is developed using React Native with Expo and serves as the presentation layer for all user roles. The backend processing layer is implemented using Node.js with Express.js and exposes REST API endpoints for authentication, profile management, job posting management, PESO referral/application handling, notification records, OCR-assisted NSRP processing, and administrative functions. The database layer uses MySQL to store structured system records, while uploaded NSRP form images are handled as file references for OCR-assisted profile encoding. The Job Seeker module allows users to register, create an NSRP-based profile, optionally use OCR-assisted NSRP form auto-fill, browse job vacancies, view rule-based skill alignment, view employer-provided application instructions, submit an application/request PESO referral, and track application/referral status. In this architecture, application submission and tracking refer to PESO-Link application/referral records, while employer-email applications remain external. All OCR-extracted data must be reviewed, edited if necessary, and confirmed by the user before being saved.

The Employer module allows administrator-approved employers to manage job postings, provide application email or instructions, and view submitted applications within the limits of the system workflow. In this architecture, submitted applications refer to PESO-Link application/referral records, while employer-email applications remain outside the system’s full tracking capability. Employers do not perform self-registration because employer accounts are created and managed under PESO administrative control.

The PESO Admin module allows PESO personnel to manage employer accounts, monitor job postings, oversee PESO referral routing, manage job seeker accounts, and support referral status tracking. Administrative functions are designed to support existing PESO procedures and do not replace official PESO validation or hiring authority.

The database stores user accounts, NSRP-based profile details, job posting records, PESO referral records, referral tracking updates, notification records, and skill matching references. File storage is used only for uploaded NSRP form images and related system files required for profile encoding support.

The integration of OCR, NSRP-based digital profile encoding, rule-based skill matching, and PESO referral generation is handled through the backend processing layer and system database. The Job Seeker module allows the user to create an NSRP-based digital profile either through manual encoding or optional OCR-assisted NSRP form-to-profile auto-fill. If OCR is used, the extracted text is returned to the user for review, editing, and confirmation before any profile record is saved. Once confirmed, the profile details and confirmed skills are stored in the database. The Employer module supplies job posting details and required skills through job posting management. When a job seeker views a job posting, the backend compares the confirmed profile skills with the required skills of the selected job posting and returns the matched and missing required skills to the mobile application. If the job seeker submits a PESO referral request, the backend creates an application/referral record connected to the job seeker profile and selected job posting for PESO Admin review, routing, and status tracking. This integration remains system-supported and does not produce a numerical score, applicant ranking, recommendation, automatic screening decision, referral approval, or hiring decision.

3.4.2 Use Case Diagram

Figure 2 presents the use case diagram of PESO-Link MisOr. The diagram shows the main functions available to each user role in the system.

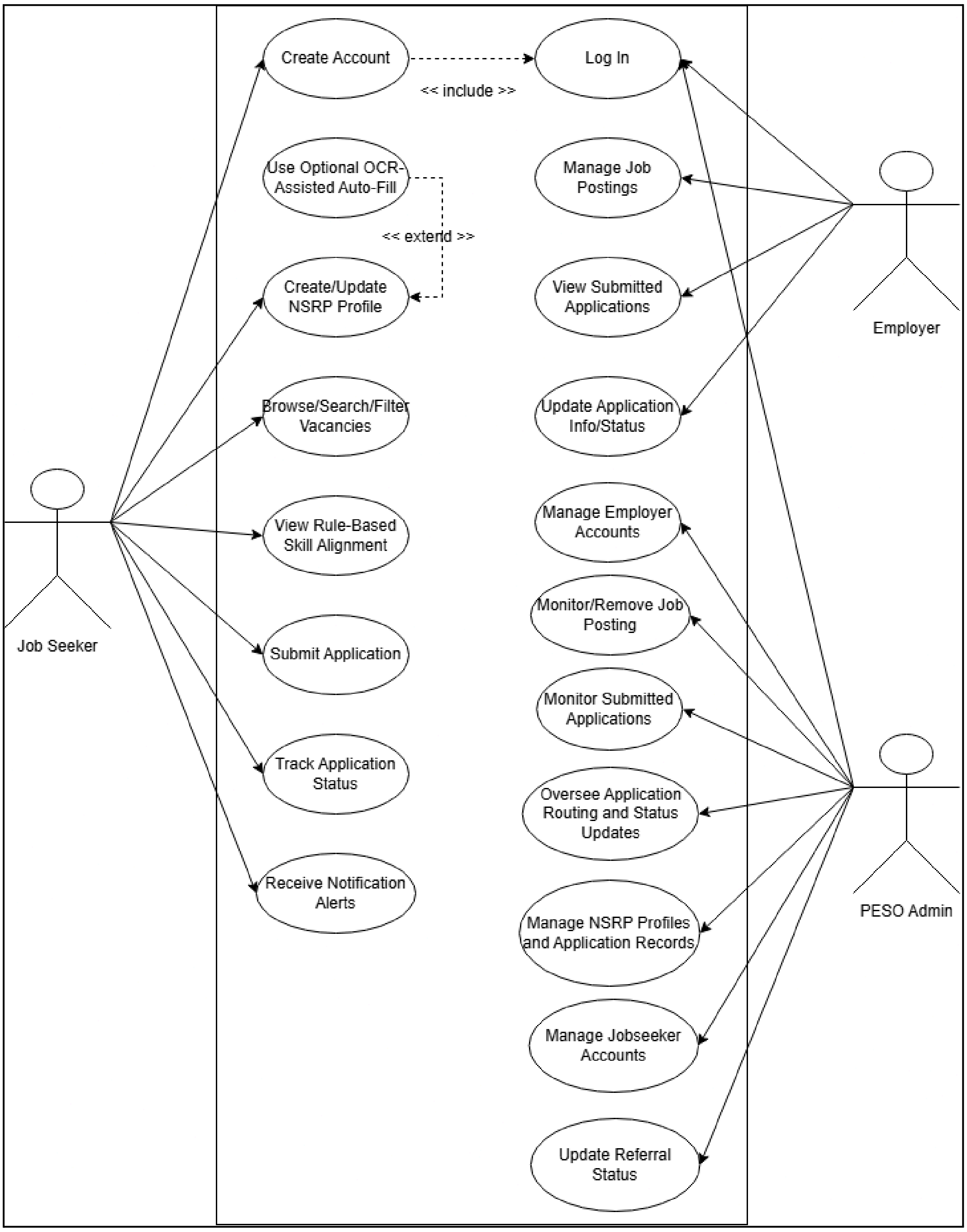

Figure 2. Use Case Diagram of PESO-Link MisOr

The Job Seeker can create an account, log in, create and update an NSRP-based profile, optionally use OCR-assisted NSRP form-to-profile auto-fill, browse job vacancies, use search and filtering functions, view rule-based skill alignment, view employer-provided application instructions, submit an application/request PESO referral, receive notification alerts, and track application/referral status.

In the context of the system workflow, application-related use cases in the diagram refer to PESO-Link referral records submitted through the system. Applications submitted through employer-provided email or external instructions are handled outside PESO-Link and are not fully tracked by the system.

The Employer can log in using an administrator-approved account, manage job postings, provide application email or instructions, view submitted applications, and update application information or status within the limits of the system workflow. In this use case diagram, submitted applications refer only to PESO-Link application/referral records submitted through the system, while applications sent through employer-provided email or external instructions remain outside the system's full tracking capability.

The PESO Admin can log in, manage employer accounts, monitor and remove job postings when necessary, manage job seeker accounts, including deactivation of invalid or incomplete accounts, monitor PESO referral requests, support referral routing, manage system records, and oversee referral status updates. These functions are designed to centralize employment facilitation activities that are currently handled through separate channels.

3.4.3 Data Flow Diagram

The Data Flow Diagram (DFD) presents how data moves between the users, system processes, and data storage of PESO-Link MisOr. For this study, the DFD is presented in two levels: the Level 0 Context Diagram and the Level 1 Data Flow Diagram.

In the data flow diagrams, application-related flows represent PESO-Link application/referral records and referral status records handled within the system. Routing refers to the movement of an application/referral record from submission to PESO Admin review and, when endorsed, to the employer within the PESO-Link workflow. Tracking refers to the storage and monitoring of status updates associated with that record after submission. Direct employer-email applications are external to PESO-Link and are therefore not represented as fully tracked internal application records.

3.4.3.1 Level 0 Context Diagram

Figure 3 presents the Level 0 Context Diagram of PESO-Link MisOr. The context diagram shows the system as one main process interacting with three external entities: Job Seeker, Employer, and PESO Admin.

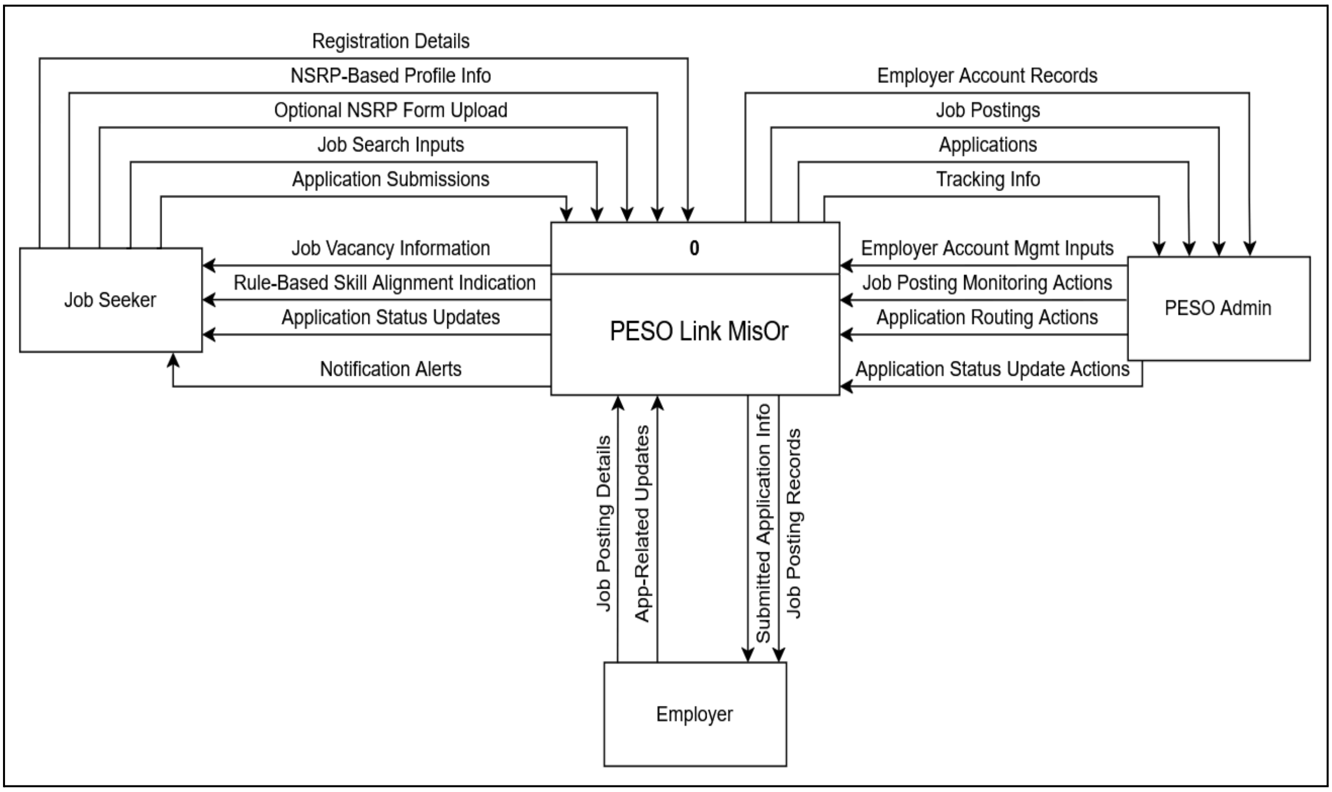

Figure 3. Level 0 Context Diagram of PESO-Link MisOr

In the Level 0 Context Diagram, the Job Seeker provides registration details, NSRP-based profile information, optional NSRP form image uploads, job search inputs, and application submissions. In this diagram, application submissions refer to PESO-Link application/referral records submitted through the system. In return, the system provides job vacancy information, employer-provided application instructions, rule-based skill alignment indication, application status updates, and notification alerts.

The Employer provides job posting details, employer-provided application instructions, and application-related status updates through an administrator-approved account. In return, the system provides job posting records and submitted application information. In this context, submitted application information refers only to PESO-Link application/referral records handled within the system, while employer-email applications remain external.

The PESO Admin provides employer account management inputs, job posting monitoring actions, application routing actions, and referral status update actions. In return, the system provides centralized records of employer accounts, job postings, application records, and tracking information. In this diagram, application records refer to PESO-Link referral records and their corresponding status updates.

The Level 0 Context Diagram shows that PESO-Link MisOr serves as a centralized mobile-based employment facilitation system while maintaining PESO administrative control over employer accounts, job posting monitoring, application routing, and application status updates.

3.4.3.2 Level 1 Data Flow Diagram

Figure 4 presents the Level 1 Data Flow Diagram of PESO-Link MisOr. The Level 1 DFD expands the main system process into its major internal processes: account management, NSRP-based profile management, optional OCR-assisted encoding, job posting management, job browsing and filtering, application submission and routing, application tracking, notification management, and rule-based skill matching. In this diagram, application submission and tracking refer to PESO-routed application/referral records handled within PESO-Link.

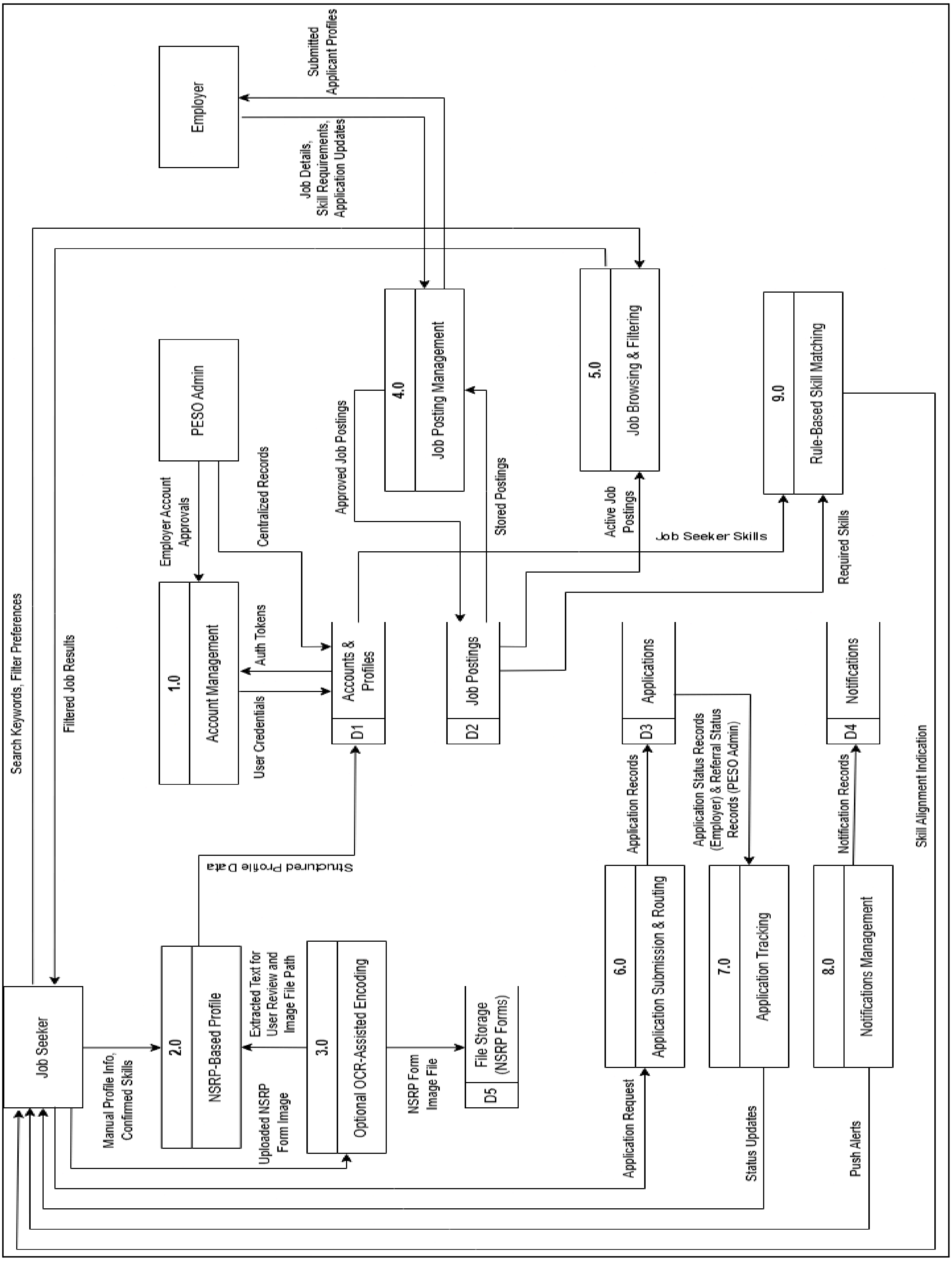

Figure 4. Level 1 Data Flow Diagram of PESO-Link MisOr

The Account Management process handles registration, login, role-based access, and employer account control. Job seekers may create accounts directly, while employer accounts are created or approved under PESO Admin control.

The NSRP-Based Profile Management process handles the creation and updating of job seeker profiles. Users may manually encode their profile information or use the optional OCR-assisted NSRP form-to-profile auto-fill process.

The Optional OCR-Assisted Encoding process handles uploaded NSRP form images. When OCR is used, the system extracts text and pre-fills selected profile fields. The extracted information must be reviewed, edited if necessary, and confirmed by the job seeker before being saved. This process does not perform automated screening, validation, ranking, or hiring decisions.

The Job Posting Management process handles the creation, updating, monitoring and management of job postings. Employers may manage job postings through approved accounts, while PESO Admin monitors job posting records in accordance with existing PESO procedures.

The Job Browsing and Filtering process allows job seekers to view localized job vacancies and search or filter postings based on available job information. This process addresses the fragmented access to job postings currently handled through Facebook and similar platforms.

The Application Submission and Routing process represents PESO-routed application/referral requests submitted through PESO-Link. It stores referral requests and connects job seeker profiles to selected job postings for PESO Admin review. This supports centralized routing among job seekers, employers, and PESO Admin. The Application Tracking process manages referral status records and allows job seekers to view updates after submitting PESO referral requests. This process directly addresses the lack of status visibility for PESO-routed applications identified in the current workflow.

The Notification Management process sends alerts related to job posting updates and PESO referral status updates. Notifications support structured updates but do not replace direct communication or official employer hiring decisions.

The Rule-Based Skill Matching process compares confirmed job seeker skills from the NSRP-based profile with the required skills listed in job postings. The output is a basic skill alignment indication. This process does not rank applicants, recommend hiring, perform automated screening, or use artificial intelligence.

The database stores structured records, including user accounts, job seeker profiles, employer accounts, job postings, PESO referral records, referral status records, notifications, and skill requirements. Uploaded NSRP form images are stored separately in file storage and are used only for optional OCR-assisted profile encoding.

3.4.4 Entity-Relationship Diagram

Figure 5 presents the Entity-Relationship Diagram of PESO-Link MisOr. The diagram shows the major data entities required to support the system.

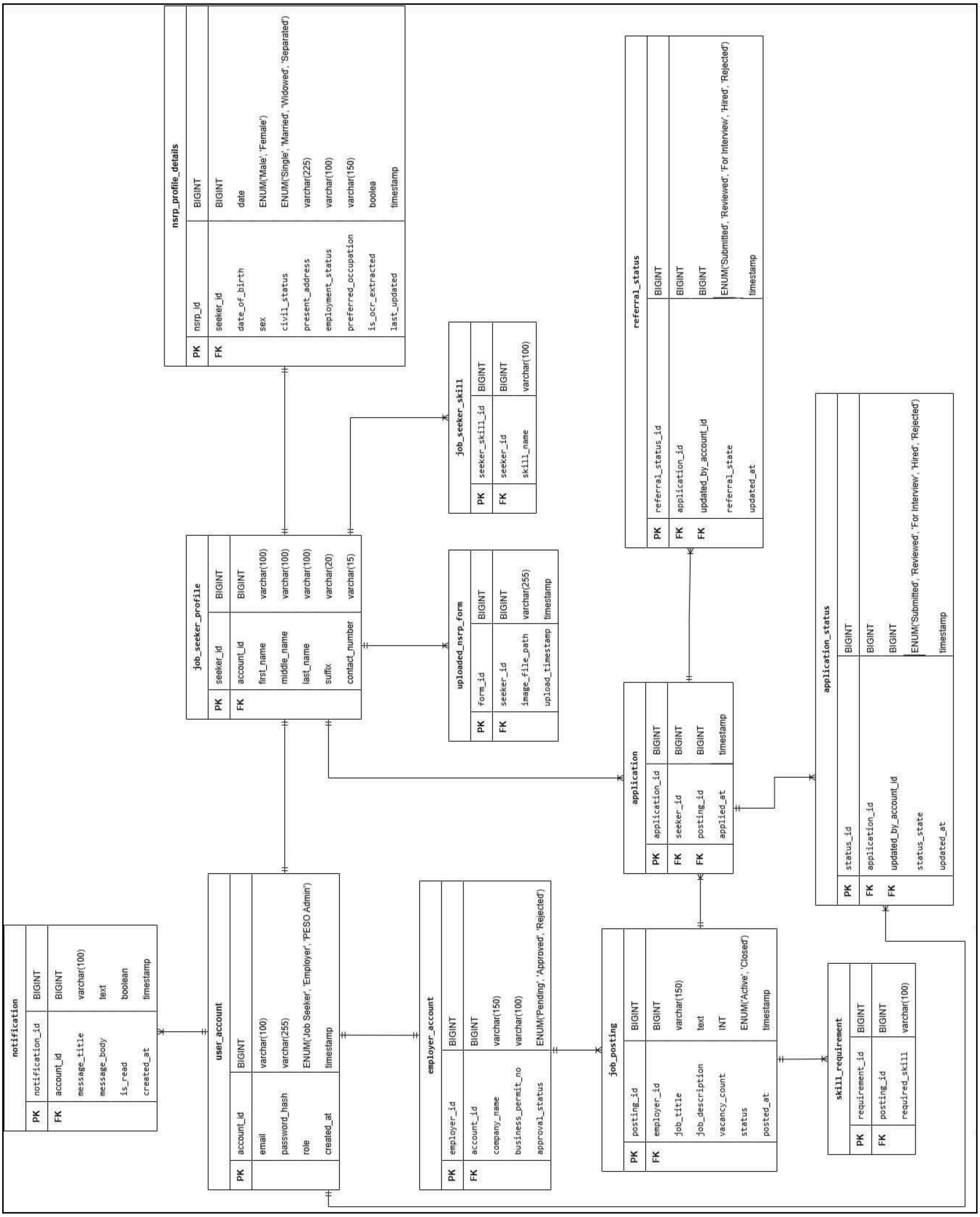

Figure 5. Entity-Relationship Diagram of PESO-Link MisOr

For consistency with the system workflow, application-related entities in the ERD refer to PESO-Link referral records and their corresponding status updates. External applications submitted through employer-provided email are not stored as fully tracked system application records.

The major entities of the system include User Account, Job Seeker Profile, Employer Account, Job Posting, Application, Application Status, Referral Status, Notification, NSRP Profile Details, Uploaded NSRP Form, and Skill Requirement.

The User Account entity stores login and role-based access information. Each user account is assigned to a role, such as Job Seeker, Employer, or PESO Admin.

The Job Seeker Profile entity stores the user's NSRP-based profile information. This includes profile details manually entered by the user or confirmed after optional OCR-assisted extraction.

The Employer Account entity stores employer information created and managed under PESO administrative control. Employer accounts are connected to job postings.

The Job Posting entity stores job vacancy information, including position details, requirements, and skills required by the employer.

The Application entity represents PESO-Link referral records submitted by job seekers. Each record connects a job seeker profile to a selected job posting for PESO Admin review.

The Application Status indicates the current processing status of the application as updated by the employer (e.g., for review, interview, hired, rejected).

The Referral Status indicates the current status of a PESO referral request for the application (e.g., pending, for review, PESO-referred, rejected).

The Notification entity stores notification records related to job posting updates and PESO referral status updates.

The Uploaded NSRP Form entity stores references to uploaded NSRP form images used only for optional OCR-assisted profile encoding.

The Skill Requirement entity stores required skills listed in job postings. These skills are compared with confirmed job seeker skills through rule-based matching to provide a basic alignment indication.

3.4.5 System Workflow

Figure 6 presents the proposed system workflow of PESO-Link MisOr. The workflow shows how job seekers, employers, and PESO administrators interact within the mobile-based employment facilitation system.

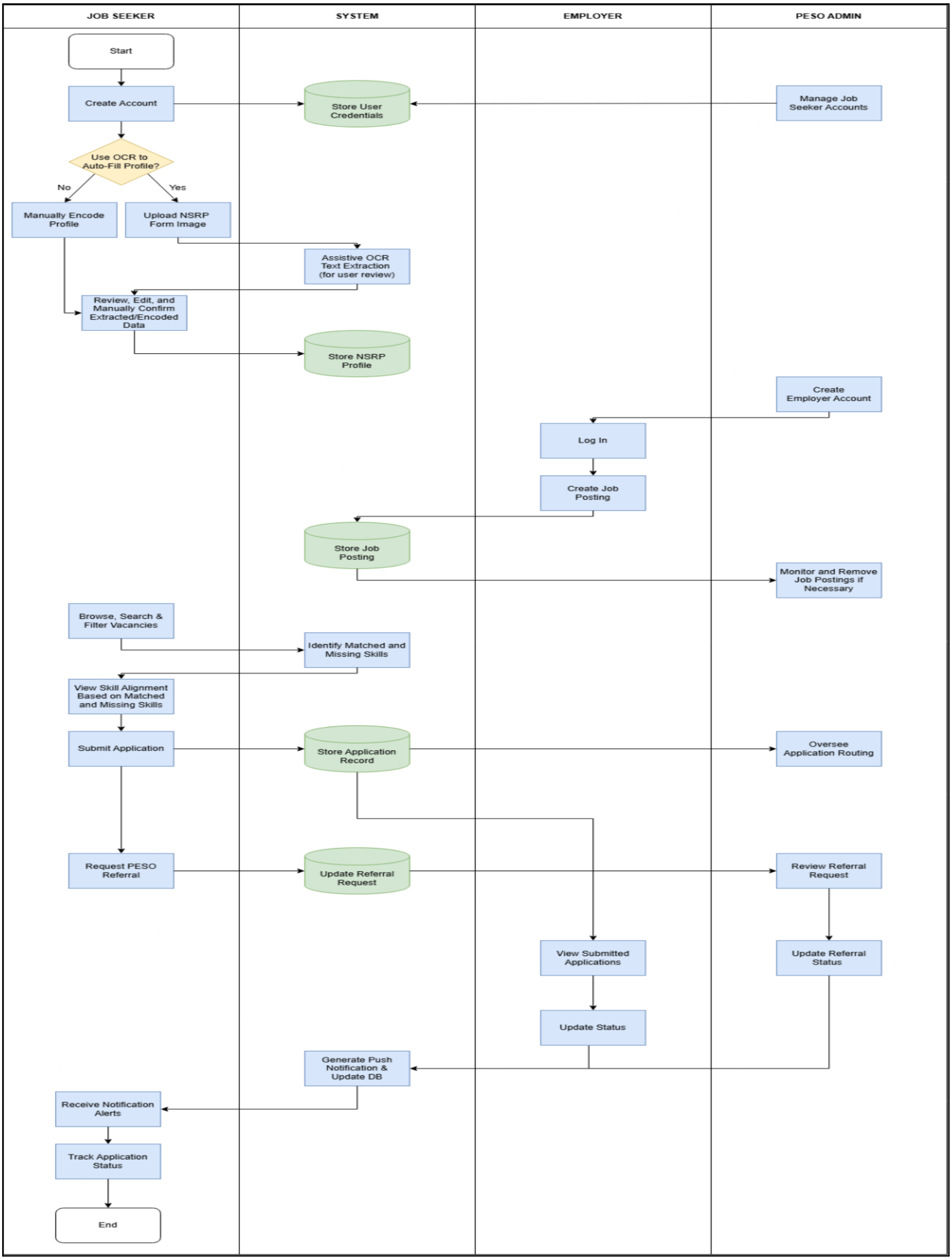

Figure 6. Proposed System Workflow of PESO-Link MisOr

The workflow begins when a job seeker creates an account and completes an NSRP-based profile. The job seeker may manually encode profile information or use the optional OCR-assisted NSRP form-to-profile auto-fill feature. When OCR is used, the job seeker uploads an image of the accomplished NSRP form. The system extracts text and pre-fills selected profile fields. The job seeker must review, edit, and confirm the extracted information before saving it to the profile.

After completing the profile, the job seeker can browse localized job vacancies using search and filtering functions. The system may display a rule-based skill alignment indication by comparing the confirmed skills in the job seeker profile with the skills required in a job posting. This matching function is limited to decision-support and does not rank applicants, recommend hiring, or make automated decisions.

When a job seeker applies through employer email, the system only displays the employer-provided email or instructions, and the application process continues outside PESO-Link. When a job seeker requests PESO referral, the system requires completion of the required NSRP-based applicant profile fields and creates a referral record for PESO Admin review. If PESO Admin endorses the request, the applicant record is marked as PESO-Referred and becomes visible to the employer. The employer may review the applicant's NSRP-based profile summary and manually update the application status. PESO referral does not guarantee hiring and does not replace employer evaluation.

Within the system workflow, skill matching occurs after the job seeker has completed or confirmed the NSRP-based profile and after job postings with required skills are available in the system. The job seeker first creates or updates the profile, either manually or with optional OCR-assisted auto-fill subject to user confirmation. The employer then creates a job posting with job details and required skills. When the job seeker browses or views a job posting, the system compares the confirmed profile skills with the job posting required skills and displays the matched and missing required skills. This output supports the job seeker's review of job requirements but does not block the user from viewing job details, following employer-provided application instructions, or requesting PESO referral. Referral review, routing, and status updates remain under PESO Admin and employer workflow authority.

For employers, system access begins with an account created or approved by PESO Admin. Once access is provided, employers can manage job postings and view submitted applications through the system. In this workflow, submitted applications refer to PESO-Link application/referral records handled within the system. Employer-side actions remain within the system workflow and do not replace employer evaluation or hiring decisions.

For PESO Admin, the workflow includes managing employer accounts, monitoring job postings, overseeing application routing, and supporting application tracking. PESO Admin functions provide centralized visibility of job posting and application activities while maintaining existing PESO validation and coordination authority.

In the PESO referral workflow, a referral request is first created when the job seeker submits a PESO referral request for a selected job posting. The system stores the request as a PESO-Link application/referral record connected to the job seeker profile and selected job posting. PESO Admin then reviews the submitted referral request and may update its status according to the processing stage, such as Submitted, For Review, PESO-Referred, Rejected, or Closed. When the status is updated, the system records the new status and generates an in-app notification alert for the job seeker. The job seeker may then view the updated referral status through the application's tracking function. This workflow provides status visibility for PESO-Link referral records while preserving PESO Admin review authority and employer hiring decision authority.

This workflow addresses the fragmented employment facilitation process by placing job vacancy browsing, application submission, application tracking, employer posting management, and PESO administrative monitoring within one mobile-based system. The workflow remains aligned with current PESO practices because the system supports coordination and tracking without replacing official PESO procedures, employer evaluation, or external government systems.

3.5 Functional Requirements

This presents the functional requirements of PESO-Link MisOr. The functional requirements were derived from the validated workflow issues identified through interviews, on-site observations, employer consultation, survey responses, and follow-up interviews. These requirements define the specific functions that the system must provide to support structured job information access, application tracking, centralized workflow management, and NSRP-based profile encoding.

The functional requirements are grouped according to the three primary user roles of the system: Job Seeker, Employer, and PESO Admin.

3.5.1 Job Seeker Functional Requirements

The Job Seeker module supports account registration, profile creation, vacancy browsing, application submission, application tracking, notification alerts, optional OCR-assisted profile encoding, and rule-based skill matching.

The system shall allow job seekers to create an account and log in using valid credentials.

The system shall allow job seekers to create and update an NSRP-based digital profile.

The system shall allow job seekers to manually encode profile information based on NSRP-related fields.

The system shall allow job seekers to optionally upload an image of an accomplished NSRP form for OCR-assisted profile auto-fill.

The system shall extract text from the uploaded NSRP form image and pre-fill selected profile fields when the OCR feature is used.

The system shall require job seekers to review, edit, and confirm OCR-extracted information before saving it to the profile.

The system shall allow job seekers to browse localized job vacancies posted within the system.

The system shall allow job seekers to search and filter job postings based on available job information.

The system shall allow job seekers to view job posting details, including position information, requirements, and employer-provided details.

The system shall provide a rule-based skill alignment indication by identifying matched and missing skills between confirmed job seeker skills and job posting requirements.

The system shall allow job seekers to submit applications to selected job postings.

The system shall allow job seekers to view the status of submitted applications.

The system shall provide notification alerts related to application status updates and newly posted job vacancies.

The system shall not allow OCR-extracted information to be saved without user confirmation.

The system shall not perform automated applicant ranking, hiring recommendations, artificial intelligence-based screening, or automated hiring decisions.

The system shall allow job seekers to view employer-provided email or application instructions for external applications.

The system shall allow job seekers to request PESO referral after completing the required NSRP-based applicant profile fields.

The system shall allow job seekers to view the status of PESO-Link referral records.

The system shall allow job seekers to view rule-based skill alignment based on matched and missing skills between their confirmed profile skills and job posting requirements. The system shall not automatically block job seekers from viewing job details, following employer-provided application instructions, or requesting PESO referral solely because of missing skills.

3.5.2 Employer Functional Requirements

The Employer module supports job posting management and applicant viewing through accounts created or approved under PESO administrative control.

The system shall allow employers to log in using an account created or approved by PESO Admin.

The system shall allow approved employers to create job postings.

The system shall allow approved employers to update job posting information within the system.

The system shall allow approved employers to view applications submitted to their job postings.

The system shall allow approved employers to view applicant profile information submitted through the application process.

The system shall allow approved employers to update the application status of PESO-Link applicant records within the limits of the system workflow, such as For Review, For Interview, Hired, or Rejected.

The system shall not allow employers to self-register without PESO Admin approval.

The system shall not replace employer evaluation, interview processes, or hiring decisions.

The system shall not include real-time messaging, interview scheduling, payment processing, or direct communication features between employers and applicants.

The system shall allow employers to include an application email or application instruction in job postings.

The system shall allow employers to view applicant records endorsed through PESO-Link, including applicant contact information, NSRP-based profile summary, matched and missing skills, application status, and referral status when applicable.

The system shall allow employers to identify applicants marked as PESO-Referred after PESO Admin endorsement.

3.5.3 PESO Admin Functional Requirements

The PESO Admin module supports administrative control, employer account management, job posting monitoring, PESO referral routing, and referral status monitoring.

The system shall allow PESO Admin to log in using authorized admin credentials.

The system shall allow PESO Admin to create, approve, update, or manage employer accounts.

The system shall allow PESO Admin to monitor job postings submitted or managed by employers.

The system shall allow PESO Admin to oversee PESO referral records submitted by job seekers.

The system shall allow PESO Admin to support PESO referral routing between job seekers and employers.

The system shall allow PESO Admin to manage or monitor referral status updates within the system workflow.

The system shall allow PESO Admin to access centralized records of job seekers, employers, job postings, PESO referral records, and referral statuses.

The system shall allow PESO Admin to monitor the use of NSRP-based profile information within the system.

The system shall allow PESO Admin to monitor job seeker accounts and deactivate accounts with incomplete, invalid, or inappropriate information when necessary.

The system shall maintain PESO administrative control over employer access and job posting monitoring.

The system shall not replace official PESO validation procedures, NSRP processing, PEIS or PhilJobNet encoding, or external government system operations.

The system shall allow PESO Admin to review PESO referral requests and update referral status as Submitted, For Review, PESO-Referred, For Interview, Hired, Rejected, or Closed.

The system shall allow PESO Admin to mark endorsed applicant records as PESO-Referred for employer visibility.

3.5.4 System-Level Functional Requirements

The system-level requirements define the functions that support the overall operation of PESO-Link MisOr across all user roles.

The system shall provide role-based access for Job Seeker, Employer, and PESO Admin users.

The system shall store user account records, job seeker profiles, employer records, job postings, PESO referral records, referral status records, notification records, uploaded NSRP form references, and skill requirement records.

The system shall provide centralized storage of job posting and PESO referral information to reduce dependence on fragmented channels.

The system shall support referral status tracking to provide job seekers with visibility after submitting PESO referral requests.

The system shall support in-app notification alerts related to PESO referral status updates and job posting updates. Referral-related notification alerts shall be generated when an authorized user updates the status of a PESO-Link referral record.

The system shall support optional OCR-assisted NSRP form-to-profile auto-fill only as an assistive encoding feature.

The system shall support rule-based skill matching only by comparing confirmed job seeker skills with job posting required skills and displaying matched and missing required skills.

The system shall allow skill matching only after the job seeker's profile skills have been confirmed.

The system shall not use artificial intelligence, machine learning, predictive analytics, automated applicant ranking, or automated hiring recommendations.

The system shall not include functions outside the study scope, such as real-time messaging, interview scheduling, payment processing, GPS tracking, biometric identity verification, or direct integration with external government databases.

3.5.5 Summary of Functional Requirements

The functional requirements of PESO-Link MisOr are designed to directly address the validated problems identified in the current PESO Misamis Oriental workflow. Structured vacancy browsing responds to fragmented job posting access. PESO referral request submission, routing, tracking, and notification alerts respond to the absence of centralized status visibility for PESO-routed applications. Employer account management, job posting monitoring, and administrative control respond to the manual and decentralized coordination workflow. NSRP-based profile encoding with optional OCR-assisted data extraction responds to the repetitive manual encoding of job seeker information.

These requirements ensure that the system remains aligned with the stated problems, objectives, scope, and limitations of the study. The system supports employment facilitation processes but does not automate hiring decisions, replace PESO authority, or claim employment outcome improvement beyond the functions directly implemented and evaluated.

3.6 Software and Hardware Requirements

This section presents the software and hardware requirements needed for the development, testing, and evaluation of PESO-Link MisOr. The listed requirements are limited to the components necessary for the Android-based mobile employment facilitation system, including account management, NSRP-based profile encoding, optional OCR-assisted profile auto-fill, job vacancy browsing, PESO referral routing and tracking, notifications, employer account management, job posting management, and administrative monitoring.

3.6.1 Software Requirements

Table 2 presents the software requirements for the development and operation of PESO-Link MisOr.

System Tier / Component	Requirement / Specification	Description
Operating System	Android OS	Required because the submitted system is limited to Android-focused mobile application deployment and testing.
Presentation Tier	React Native with Expo	Used to develop the mobile application interface for Job Seeker, Employer, and PESO Admin users. Expo supports the Android-focused development workflow used in the prototype while allowing faster mobile interface testing during development.
Frontend Navigation	Expo Router and React Native navigation components	Used to organize mobile screens, role-based navigation flows, and user interface routing within the application.
Frontend API Communication	Axios / REST API requests	Used by the mobile application to send and receive JSON-based data from the backend API for login, profiles, job postings, referral requests, notifications, and status updates.
Application Tier / Backend	Node.js with Express.js	Used to implement the backend server and REST API routes that process authentication, user roles, NSRP profile

System Tier / Component	Requirement / Specification	Description
		records, OCR-related requests, job postings, PESO referral records, application status updates, and administrative functions.
Database / Data Tier	MySQL	Used to store structured system records such as user accounts, job seeker profiles, employer records, job postings, job seeker skills, job requirements, PESO referral/application records, status history, notification records, and uploaded NSRP form references.
Database Connector	MySQL2	Used by the Node.js backend to connect to and query the MySQL database.
Authentication and Security Support	JSON Web Token, bcryptjs, dotenv, and CORS	Used to support role-based authentication, password hashing, environment variable configuration, and controlled API communication between the mobile frontend and backend server.
Image Upload Support	Expo Image Picker and backend	Used to support selecting or uploading NSRP form images for optional OCR-assisted profile encoding during development and testing.

System Tier / Component	Requirement / Specification	Description
	file/image handling	
OCR Processing Tool	Tesseract.js	Used to support optional OCR-assisted extraction of text from uploaded NSRP form images. The extracted data remains subject to user review, editing, and confirmation before saving.
Development and Testing Tools	Node.js, npm, Expo development tools, Android emulator or connected Android device, and API testing tools	Used to install dependencies, run the backend server, start the Expo Android development flow, test API routes, and verify mobile application functions during development.
Version Control Tool	Git and GitHub	Used to manage source code, track development changes, and store the project repository for collaboration and documentation.
Notification Service	Backend notification records and	Used to support notification alerts for job posting updates, PESO referral status updates, account-related updates,

System Tier / Component	Requirement / Specification	Description
	in-app notification screens	and application status updates. Notification records are stored and retrieved through the Node.js Express backend and MySQL database and displayed within the application's notification screens.
Diagramming / Prototyping Tool	Figma, Draw.io, or equivalent tool	Used to prepare prototype screens, system architecture, use case diagram, DFD, ERD, and system workflow.
Testing and Evaluation Tool	Google Forms and spreadsheet software	Used to collect and summarize functionality and usability evaluation results.

3.6.2 Hardware Requirements

Table 3 presents the hardware requirements for the development and operation of PESO-Link MisOr.

Hardware Component	Requirement / Specification	Description

Development Computer / Laptop	Laptop or desktop computer capable of running Node.js, npm, Expo development tools, MySQL, and an Android emulator or connected Android device	Used for coding, backend setup, database configuration, mobile frontend testing, debugging, documentation, and preparing system files.
Android Smartphone	Android device compatible with the target application version	Used for installing, running, and testing the mobile-native PESO-Link MisOr application.
Device Camera	Built-in Android smartphone camera	Used to capture or upload NSRP form images for optional OCR-assisted profile encoding.
Internet Connection	Stable internet connection	Required for backend/database communication, account access, job posting updates, referral status updates, notification services, and testing.
Storage Capacity	Sufficient device and server/cloud storage capacity	Used to store application files, user records, job posting data, notification records, and uploaded NSRP form image references.

Backup Testing Device	Additional Android device, if available	Used to check basic compatibility, screen responsiveness, and usability across more than one device.

3.6.3 Technical Scope of the System

Technically, PESO-Link MisOr is implemented as an Android-focused mobile application using React Native with Expo for the presentation layer, Node.js with Express.js for the backend API layer, and MySQL for structured data storage. The mobile frontend communicates with the backend through REST API requests. The backend handles authentication, role-based access, NSRP profile records, job posting records, PESO referral/application records, status updates, in-app notification records, and OCR-related processing. OCR support is implemented only as an optional assistive feature for extracting text from uploaded NSRP form images, and all extracted information remains subject to user review and confirmation.

PESO-Link MisOr will be developed as a mobile-native Android application. The system will support role-based access for Job Seekers, Employers, and PESO Admin users. It will include the core functions defined in the scope of the study, including job seeker registration, NSRP-based profile creation, optional OCR-assisted NSRP form-to-profile auto-fill, localized vacancy browsing, search and filtering, application submission and tracking, notification alerts, employer account management, job posting management, rule-based skill matching, and PESO administrative monitoring.

The system will not include iOS deployment, web-based deployment, real-time messaging, interview scheduling, payment processing, GPS tracking, biometric identity verification, artificial intelligence-based applicant screening, automated hiring recommendations, automated applicant ranking, or integration with external government databases such as PEIS or PhilJobNet.

The technical requirements are therefore aligned with the system's stated scope and limitations. They support the development and evaluation of the mobile-based employment facilitation system without expanding the study beyond the validated PESO workflow problems.

3.7 Research Instrument

This section presents the instruments that will be used to gather data during system testing and evaluation. The instruments are limited to measuring the functionality and usability of PESO-Link MisOr based on the system's stated objectives, scope, and functional requirements.

The study will use a functional testing checklist and a usability evaluation questionnaire. These instruments will be used to determine whether the system functions operate as intended and whether users can perform assigned tasks using the system prototype.

The research instruments are grouped into two categories: a functional testing checklist and a usability evaluation questionnaire.

3.7.1 Functional Testing Checklist

The functional testing checklist will be used to verify whether the system features work according to the defined functional requirements. Each function will be tested using prepared test cases. The checklist will include the module or feature tested, test action, expected output, actual output, and result.

The result of each test case will be marked as Passed or Failed. A function will be marked as Passed if the actual output matches the expected output. A function will be marked as Failed if the actual output does not match the expected output or if the function does not operate according to the stated requirement.

The functional testing checklist will cover the following system modules: account registration and login, role-based access, NSRP-based profile creation, optional OCR-assisted NSRP form-to-profile auto-fill, job vacancy browsing, search and filtering, job posting management, application submission and routing, application tracking, notification alerts, employer account management, PESO administrative monitoring, and rule-based skill alignment indication.

3.7.2 Usability Evaluation Questionnaire

The usability evaluation questionnaire will be used to assess the users' perception of the system after interacting with the prototype. The questionnaire will focus on the usability of the system interface and functions, including ease of navigation, clarity of information, task completion, and overall system use.

The questionnaire will be answered by selected participants after they perform assigned tasks based on their user role. Job seeker participants will evaluate functions related to registration, profile creation, job browsing, application submission, tracking, notifications, optional OCR-assisted encoding, and skill alignment viewing. Employer participants will evaluate functions related to job posting management and applicant viewing. PESO Admin participants will evaluate functions related to employer account management, job posting monitoring, application monitoring, and status tracking.

The usability questionnaire will not be used to measure employment success, hiring outcomes, job placement improvement, labor market impact, or long-term deployment effectiveness. It will only measure user perception of the system's usability during prototype testing.

3.7.3 Validation of Instruments

The research instruments will be reviewed to ensure that the checklist and questionnaire items are aligned with the system's objectives, scope, and functional requirements. Items that measure features outside the system scope, such as automated hiring, applicant ranking, real-time messaging, interview scheduling, payment processing, GPS tracking, biometric verification, or external government database integration, will not be included.

The instruments will focus only on the implemented and testable components of PESO-Link MisOr. This ensures that the evaluation remains aligned with the validated problems and does not introduce unsupported claims.

3.8 Testing and Evaluation Procedure

This section presents the testing and evaluation procedure for PESO-Link MisOr. The procedure is limited to verifying whether the system functions according to its stated requirements and determining the usability of the system based on user interaction with the prototype. The evaluation will not measure employment success, hiring outcomes, labor market impact, or long-term deployment effectiveness.

3.8.1 Functional Testing

Functional testing will be conducted to determine whether the core features of PESO-Link MisOr operate according to the defined functional requirements. The researchers will prepare test cases for each major system module and record whether each function passes or fails based on expected system behavior.

The functional testing will cover the following modules:

1. Account registration and login
2. Role-based access for Job Seeker, Employer, and PESO Admin
3. NSRP-based profile creation and updating
4. Optional OCR-assisted NSRP form-to-profile auto-fill
5. Job vacancy posting and management
6. Job vacancy browsing, searching, and filtering
7. Rule-based skill alignment indication
8. PESO referral request submission and routing
9. Referral status tracking
10. Notification alerts
11. Employer account management under PESO Admin control
12. PESO administrative monitoring, including job seeker account monitoring and deactivation

Each test case will include the function to be tested, input or action performed, expected output, actual output, and result. A function will be marked as passed if the actual output matches the expected output. A function will be marked as failed if the system does not produce the expected output or if the function does not operate according to the stated requirement.

3.8.2 Usability Evaluation

Usability evaluation will be conducted after functional testing. Selected participants will use the system prototype based on assigned user roles. Job seeker participants will test account creation, profile creation, vacancy browsing, employer-provided application instruction viewing, PESO referral request submission, referral status tracking, notification viewing, optional OCR-assisted encoding, and skill alignment viewing. Employer participants will test employer login, job posting management, and applicant viewing. PESO Admin participants will test employer account management, job posting monitoring, PESO referral request monitoring, and referral status tracking.

After completing the assigned tasks, participants will answer the usability evaluation questionnaire. The usability evaluation will measure the participants' perception of the system's ease of use, clarity of navigation, task completion, and overall system use.

The usability evaluation will not be used to claim that the system improves employment outcomes, increases hiring success, improves hiring decisions, or reduces labor market problems. It will only determine whether the participants find the system usable based on their interaction with the prototype.

3.8.3 Scope of Testing

The testing will focus only on the implemented features of PESO-Link MisOr. These include account management, NSRP-based profile encoding, optional OCR-assisted profile auto-fill, job vacancy browsing, job posting management, employer-provided application instruction viewing, PESO referral request submission and routing, referral status tracking, notification alerts, employer account management, PESO administrative monitoring, and rule-based skill alignment indication.

The testing will not include features outside the scope of the study, such as iOS deployment, web-based deployment, real-time messaging, interview scheduling, payment processing, GPS tracking, biometric verification, artificial intelligence-based applicant screening, automated hiring recommendations, or integration with external government systems.

3.9 Gantt Chart

The project timeline presents the planned sequence of activities for the development of PESO-Link MisOr from software development preparation up to final defense preparation. The timeline follows the Agile development approach used in this study, where requirements analysis, system design, module development, testing, refinement, documentation, and final preparation are conducted in organized phases. The activities are aligned with the study objectives, system requirements, and evaluation procedures to ensure that the proposed mobile-based employment facilitation system is developed, tested, and documented systematically.

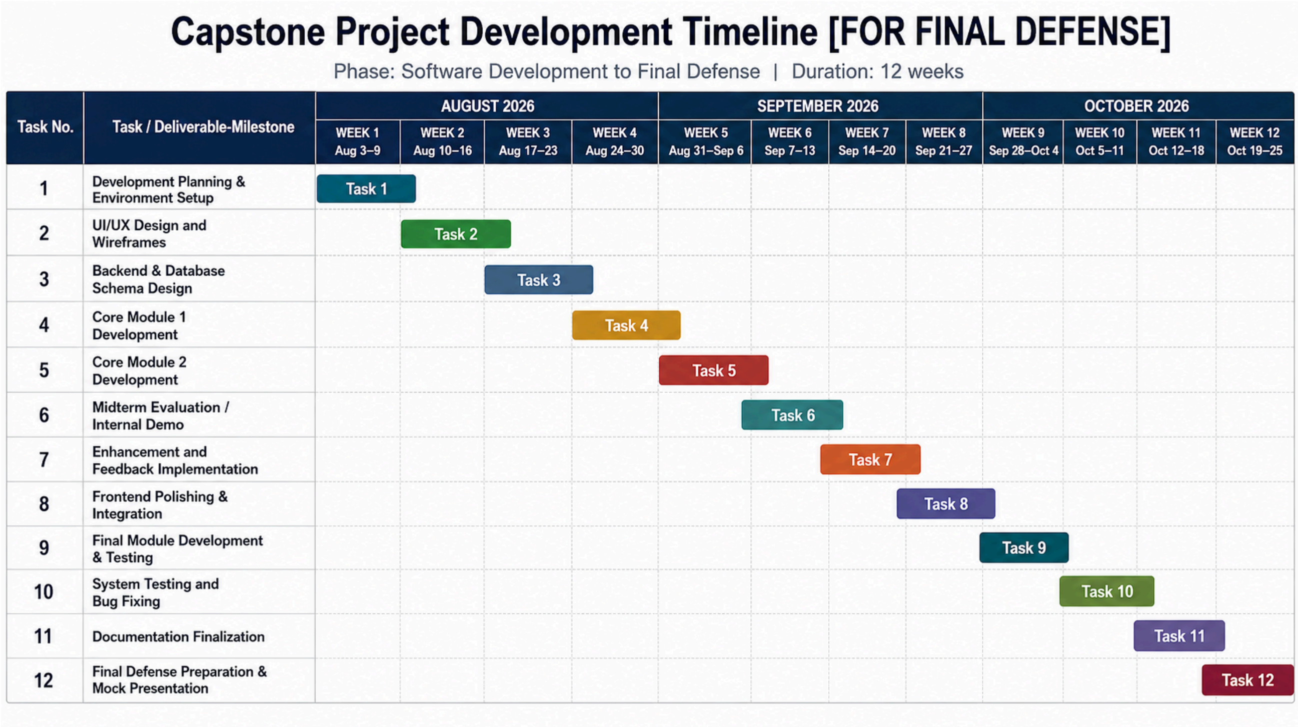

Figure 7. Gantt Chart for PESO-Link MisOr

The Gantt chart shows that the early phases focus on development planning, UI/UX design, and database/backend preparation. The middle phases focus on core module development, internal evaluation, feedback implementation, and frontend-backend integration. The final phases focus on completing the remaining system modules, including PESO referral tracking, optional OCR-assisted NSRP profile encoding, rule-based matched/missing skill alignment, notification alerts, system testing, documentation, and final defense preparation.

3.10 Ethical Considerations

This study observed and will continue to observe ethical considerations in the conduct of interviews, surveys, follow-up interviews, system testing, and usability evaluation. The researchers informed and will inform the participants about the purpose of the study and their role in the data gathering or system evaluation process before participation.

Participation in the study is voluntary. Respondents and participants were not and will not be forced to answer the survey, join follow-up interviews, or participate in system evaluation activities. They may decline or withdraw from participation without penalty.

The researchers will maintain the confidentiality of the participants' responses and personal information. Data gathered from interviews, surveys, follow-up interviews, and system evaluation will be used only for academic and research purposes. Any information presented in the study will be summarized and reported in a way that does not personally identify the participants.

For the system prototype, user information entered during testing will be used only to evaluate the functionality and usability of the system. Uploaded NSRP form images, if used during testing of the optional OCR-assisted feature, will be handled only for profile encoding evaluation and will not be used for hiring decisions, applicant screening, or external government system processing.

The researchers will also ensure that the proposed system does not replace PESO validation authority, employer evaluation, or official hiring decisions. The system will be evaluated only in terms of functionality and usability, and the study will not claim employment success, hiring improvement, or labor market impact.

3.11 Data Analysis

The data gathered from the functional testing checklist and usability evaluation questionnaire will be analyzed using descriptive methods.

For functional testing, the results will be summarized using pass or fail outcomes. The number of passed and failed test cases will be counted to determine whether the system functions operate according to the defined requirements.

For usability evaluation, the responses will be summarized using frequency, percentage, and weighted mean. These measures will be used to describe the participants' assessment of the system's usability after interacting with the prototype.

The analysis will be limited to functionality and usability results. It will not be used to claim employment success, hiring improvement, job placement increase, labor market impact, or long-term operational effectiveness.

Prototype of the PESO-Link MisOr System

The prototype of the PESO Link System presents the initial interface design of the proposed mobile-based employment facilitation system. It demonstrates the core system functionalities, including job posting, job application, applicant tracking, and administrative monitoring.

The prototype was developed using Figma to visualize the user interface and interaction flow for job seekers, employers, and PESO administrators. It reflects how users will navigate the system, access job opportunities, submit applications, and manage employment-related processes.

This prototype serves as a visual representation of the proposed system design and supports the understanding of system features and workflow prior to full system development.

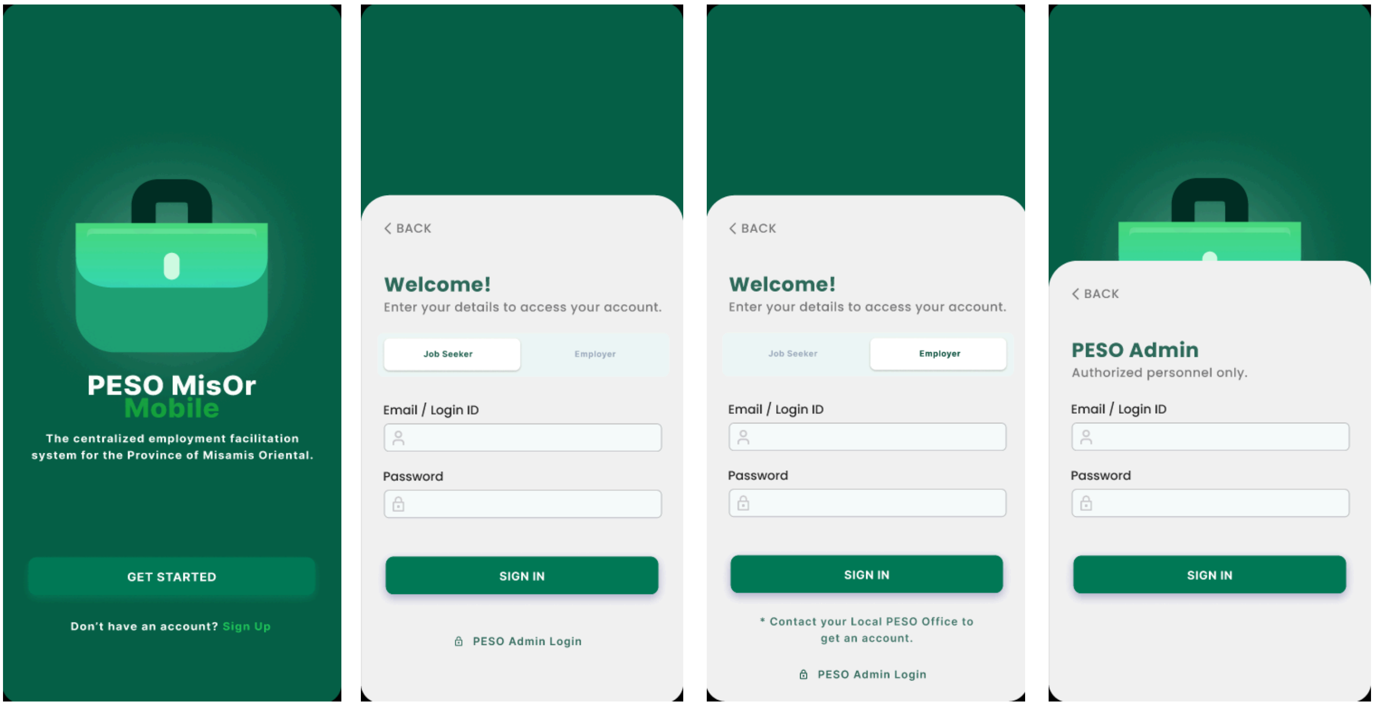

Figure 8. System Login Screens

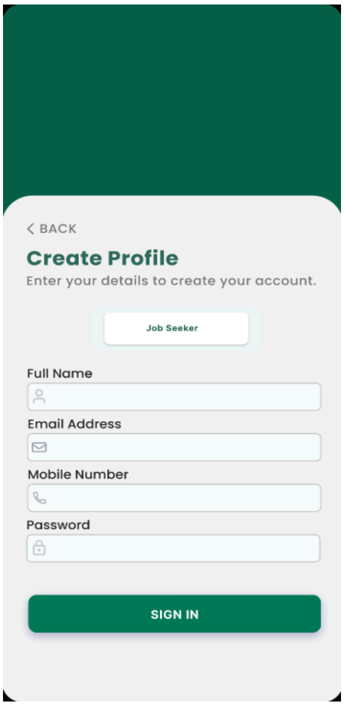

Figure 9. Job Seeker Create Account Screen

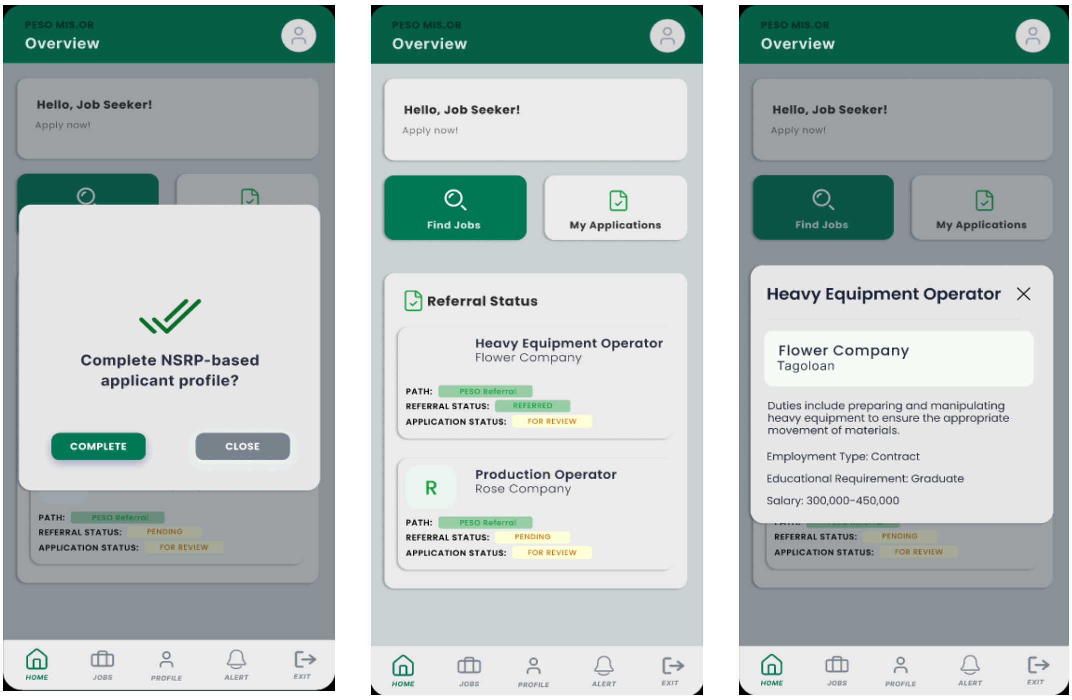

Figure 10. Job Seeker Dashboard Screens

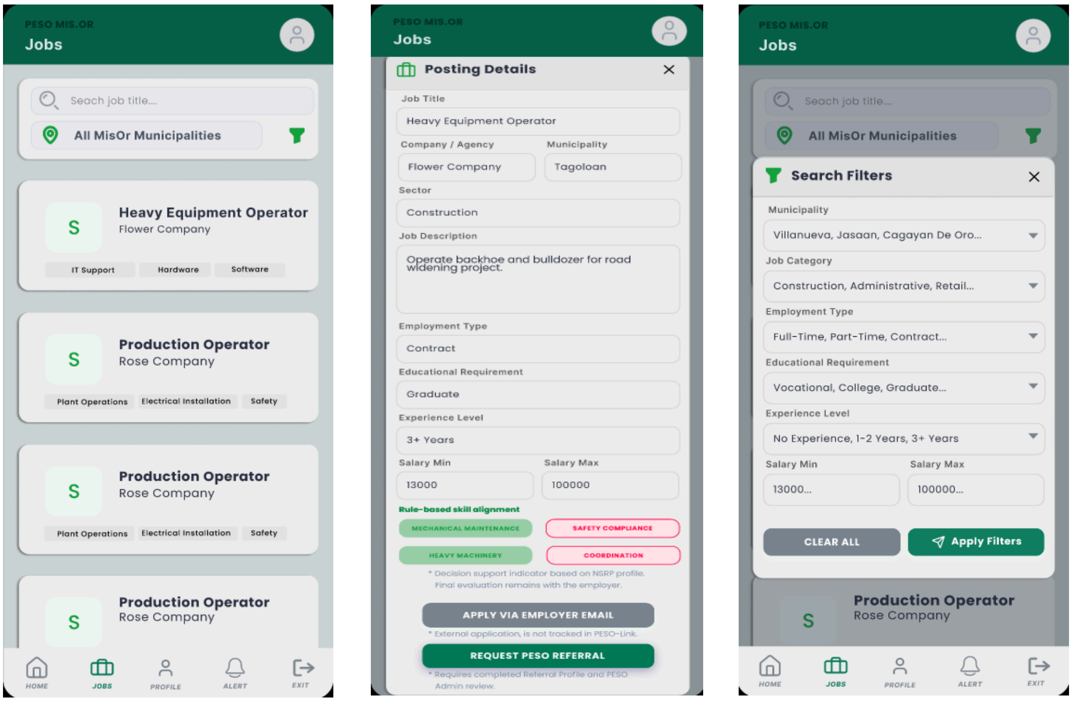

Figure 11. Job Seeker Job List, Job Details, and Filter Screens

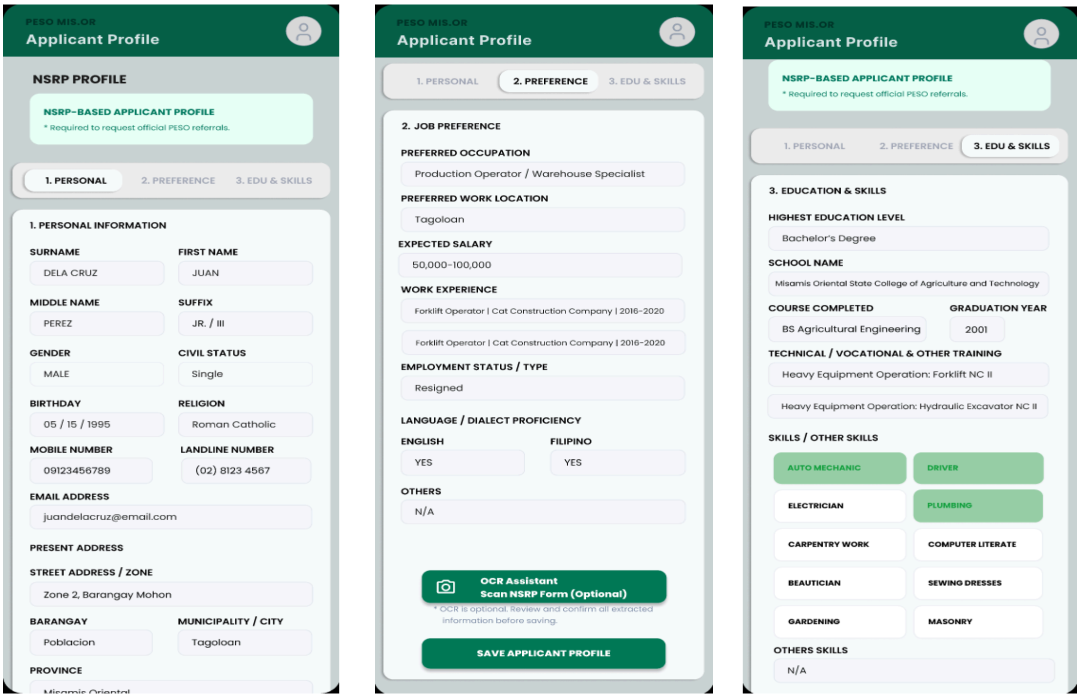

Figure 12. Job Seeker Profile Screens

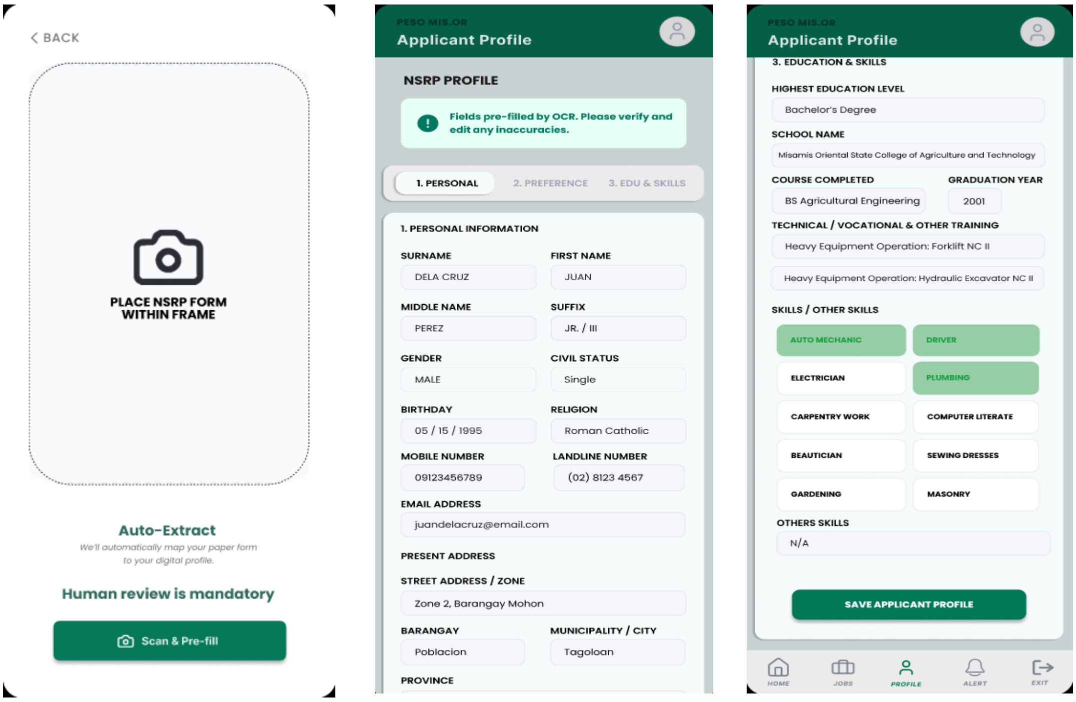

Figure 13. Job Seeker OCR Screens

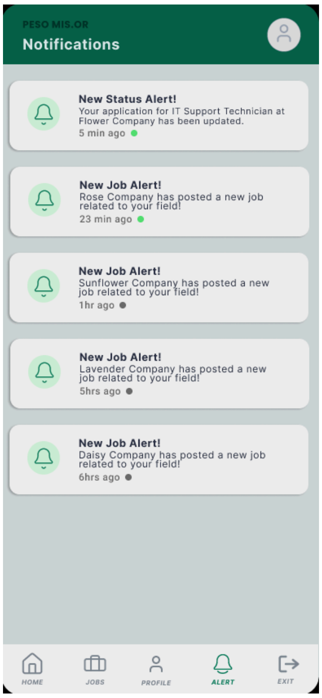

Figure 14. Job Seeker Alert Screen

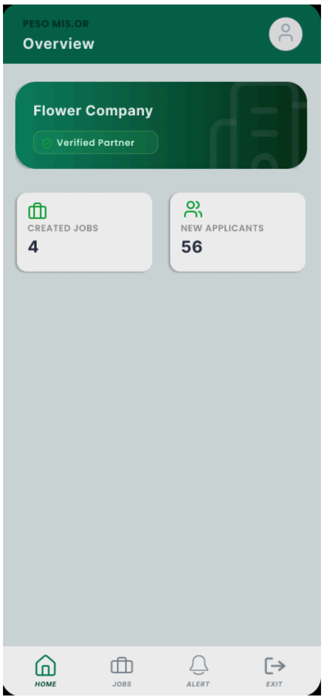

Figure 15. Employer Dashboard Screen

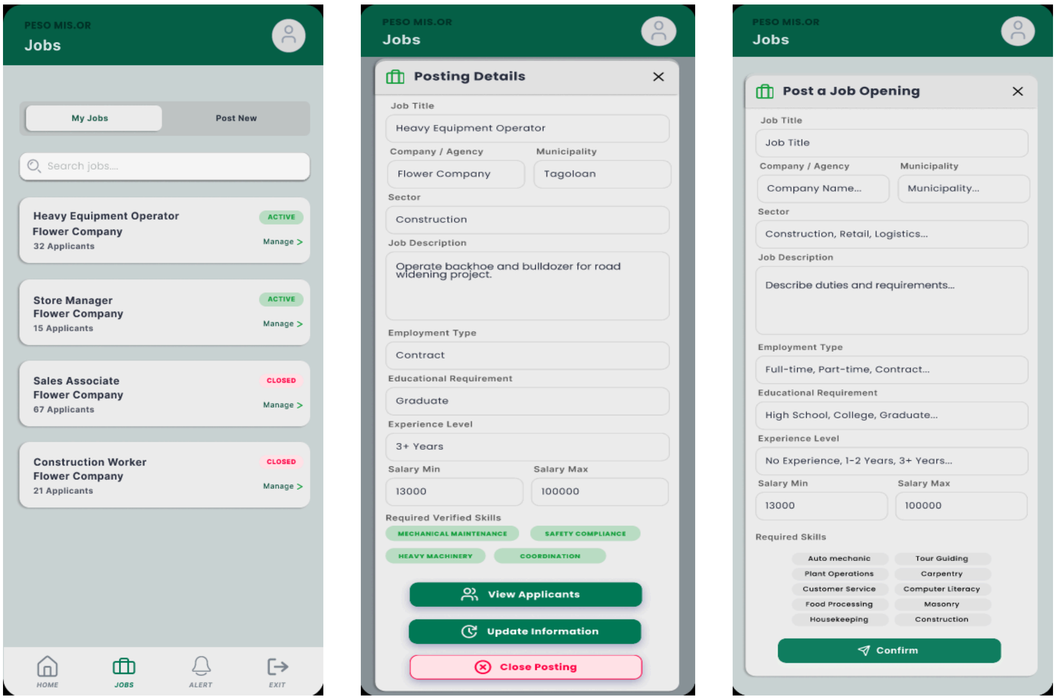

Figure 16. Employer Job List, Job Details, and Post New Job Screens

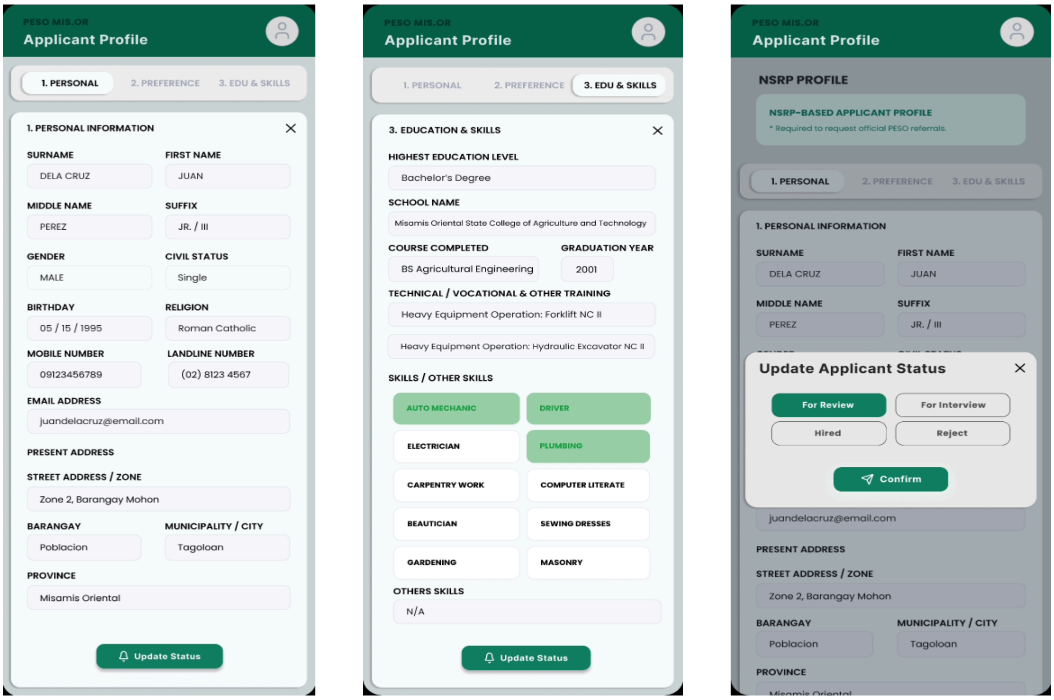

Figure 17. Employer View Applicant Profile and Update Status Screens

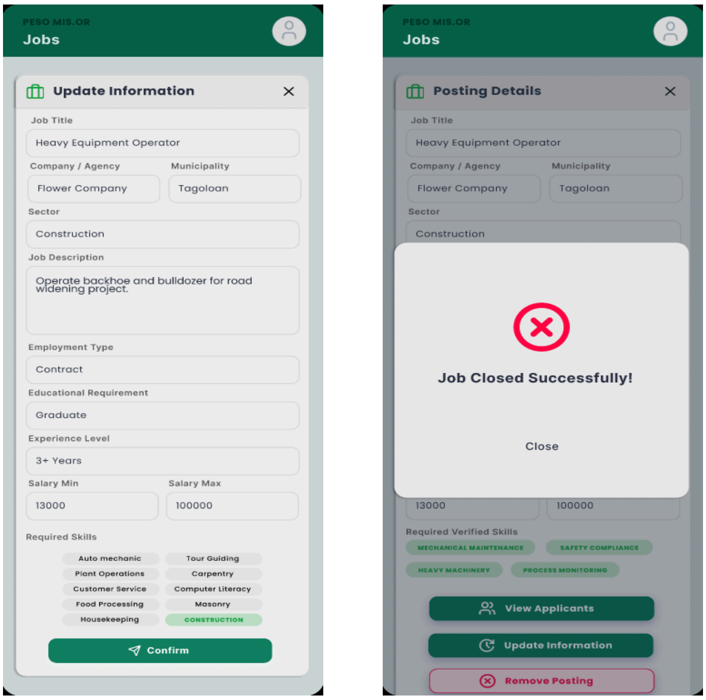

Figure 18. Employer Update Posting and Close Posting Screens

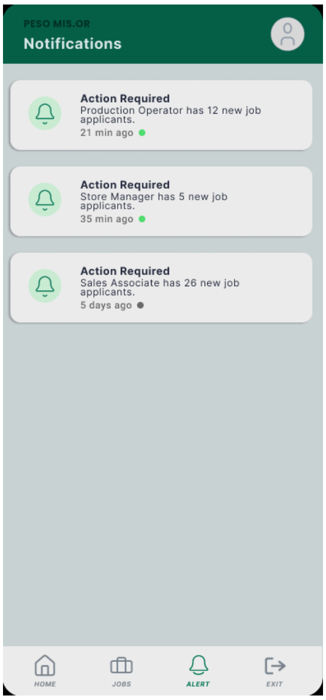

Figure 19. Employer Alert Screen

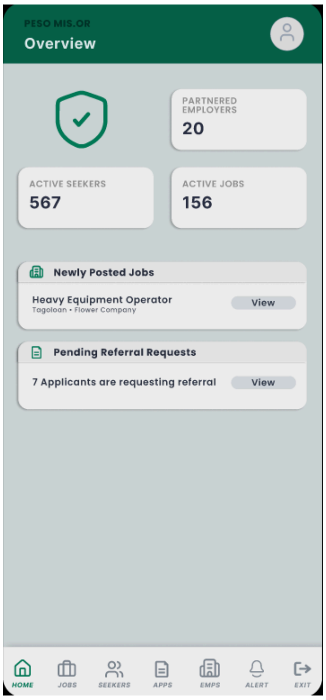

Figure 20. PESO Admin Dashboard Screen

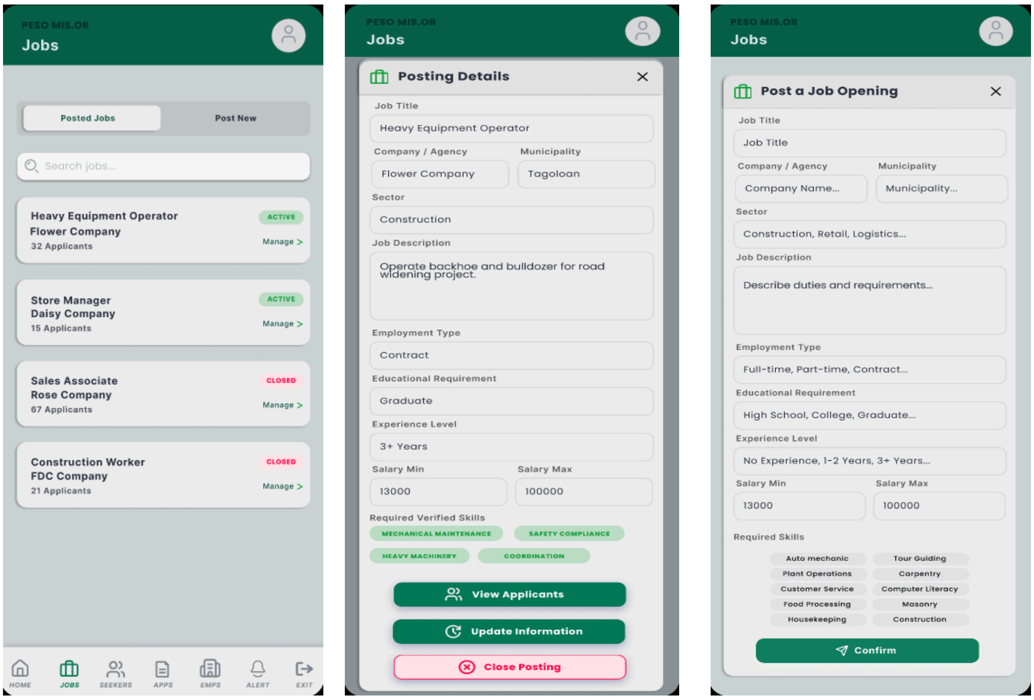

Figure 22. PESO Admin Jobs List, Job Details, and Post New Job Screens

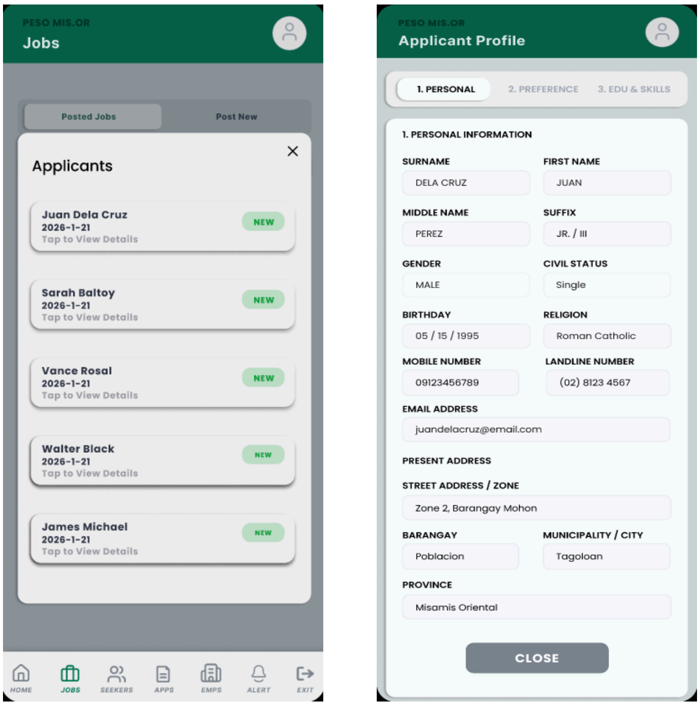

Figure 23. PESO Admin View Applicant Profile Screens

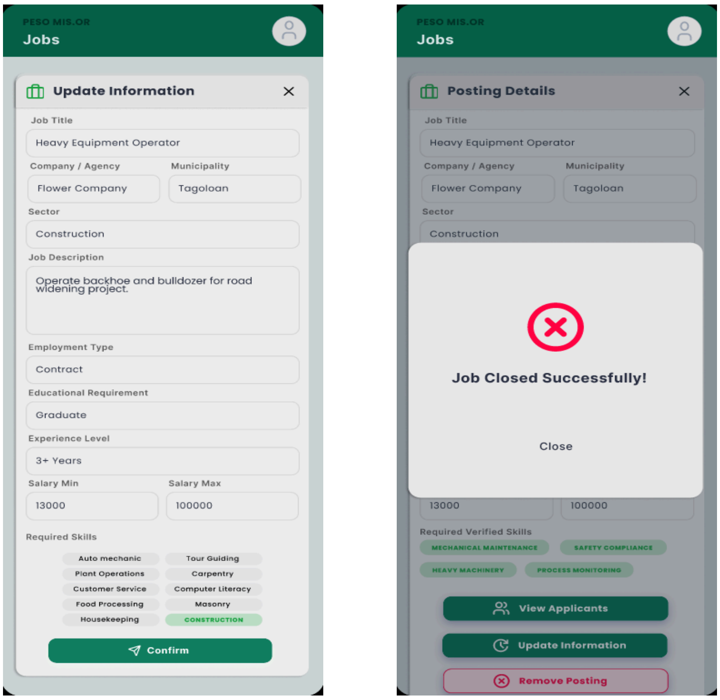

Figure 24. PESO Admin Update Posting and Close Posting Screens

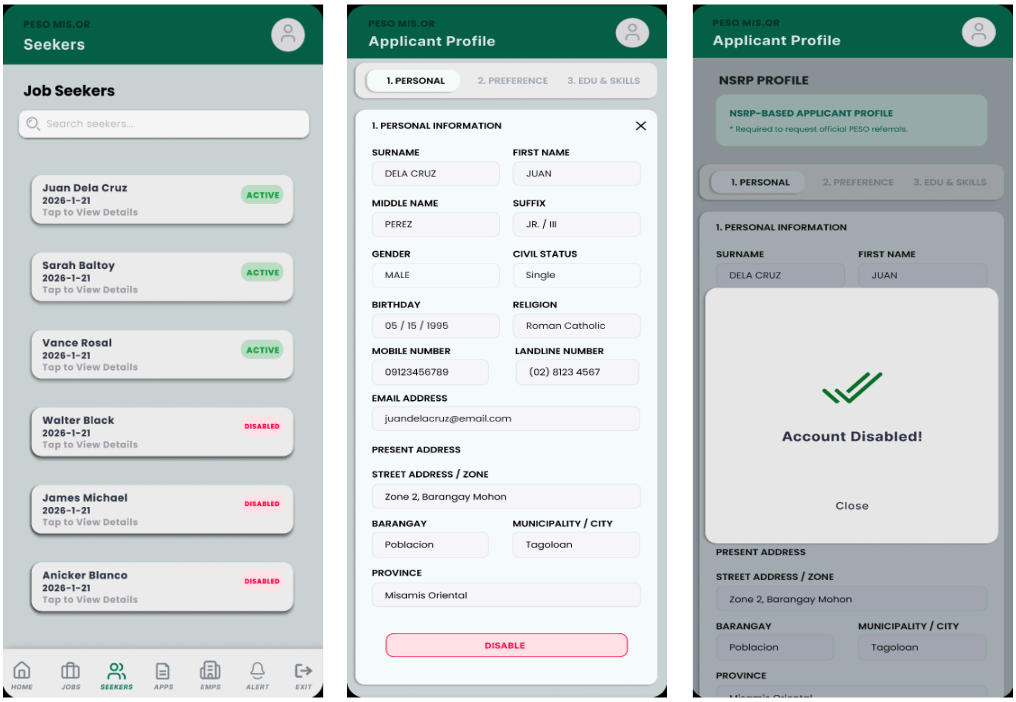

Figure 25. PESO Admin View Job Seekers Details
and Disable Job Seeker Account Screens

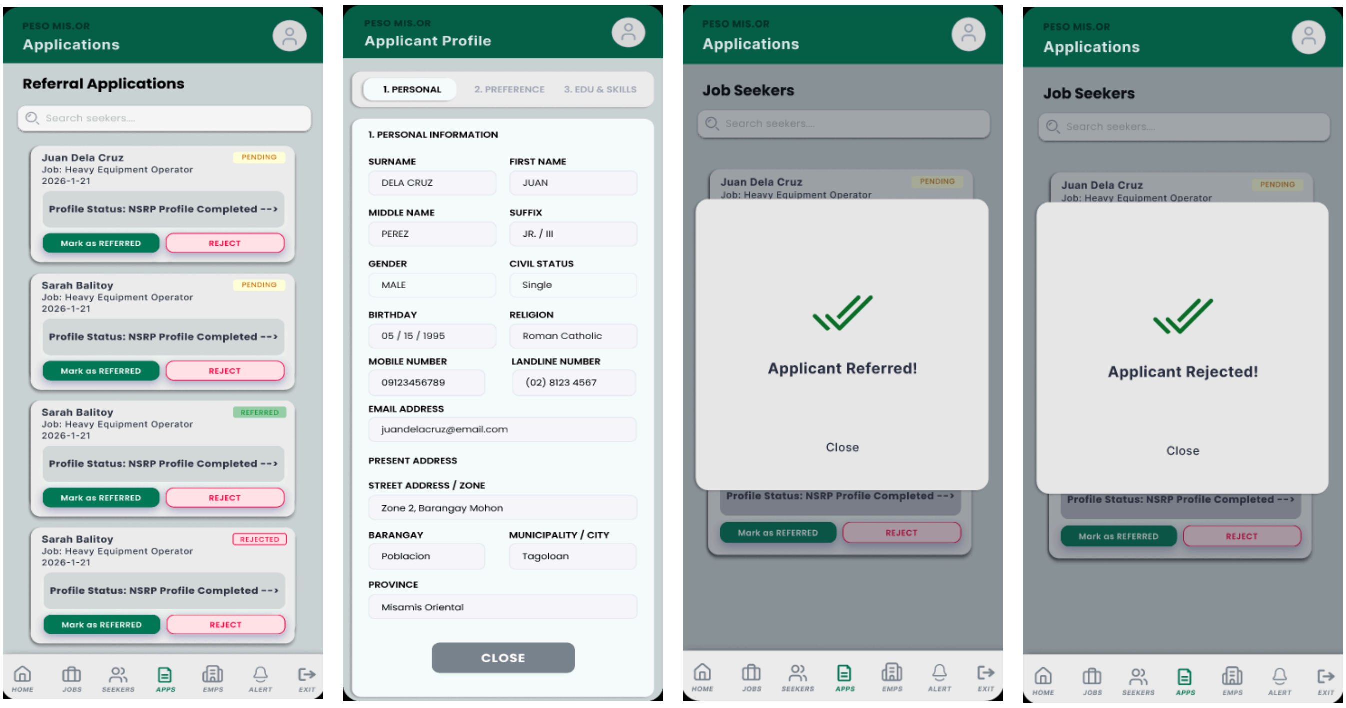

Figure 26. PESO Admin Referral Applications Screens

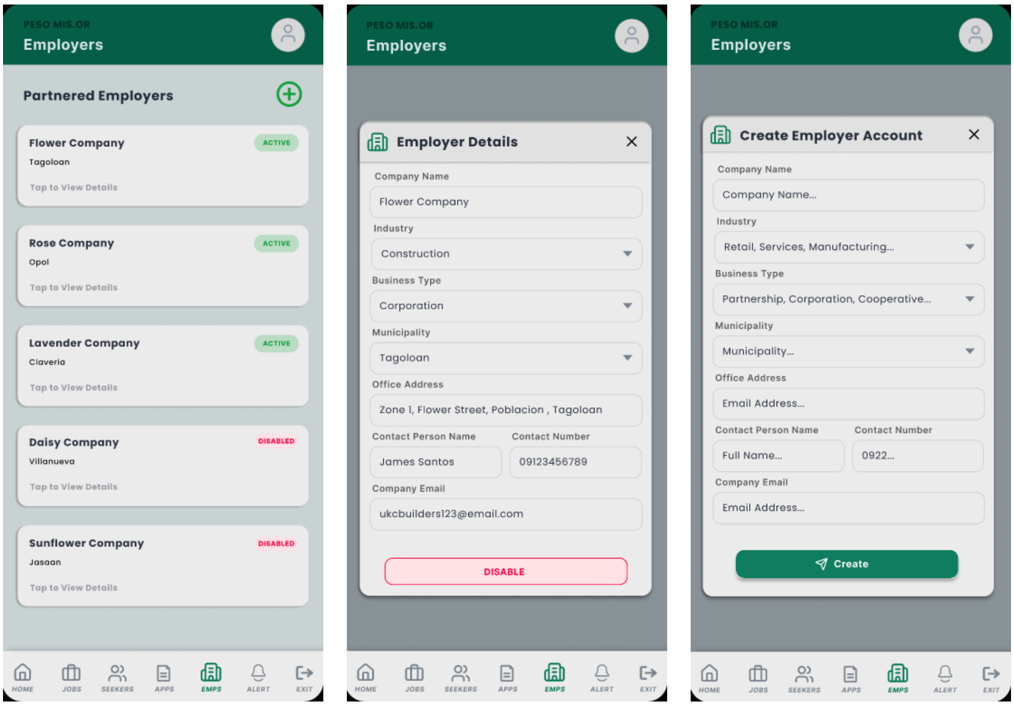

Figure 27. PESO Admin Employer List, Employer Details, Disable Employer Account, and Create Employer Accounts Screens

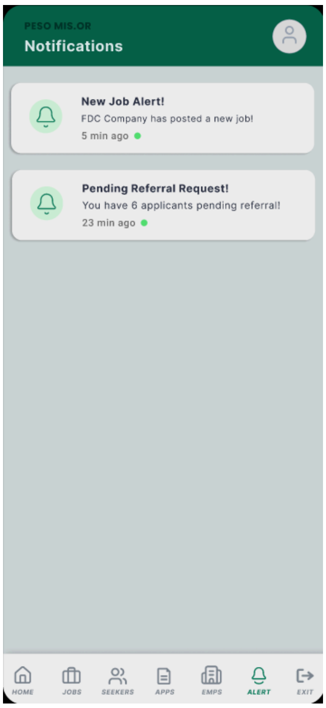

Figure 28. PESO Admin Alert Screen

Photo Documentation For Data Gathering

Figure 29. Conduct of Interviews and Consultation with PESO Misamis Oriental

Figure 30. Data Gathering Activities with Job Seeker Respondents

References

Adaza, D. N., Caneda, E. J., Tuyor, S., & Zamayla, J. (2018). Job online filtering system with background checking using web crawler [Capstone project]. College of Information Technology and Computing, University of Science and Technology of Southern Philippines.

Bacasan, J. L., Bangcal, A. M. Q., Coralde, M. R. R., & Maape, D. G. (2022). *PWDJobMatch: Web based job matching platform for persons with disabilities* [Capstone project]. College of Information Technology and Computing, University of Science and Technology of Southern Philippines.

Baluran, E., et al. (2018). Job Alert: Online job fair management system [Capstone project]. College of Information Technology and Computing, University of Science and Technology of Southern Philippines.

Brooke, J. (1996). SUS-A quick and dirty usability scale. Usability evaluation in industry 189, 194 (1996), 4–7

Davis, F.D. 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. *MIS quarterly*, *13*(3), pp.319-340. Department of Information and Communications Technology (2024). National ICT Household Survey 2024 (NICTHS): Region X Report. https://ictstatistics.dict.gov.ph/nicths2024/

Department of Labor and Employment. (n.d.). PhilJobNet. PhilJobNet. https://philjobnet.gov.ph/

Fritz.ai. (2023). Optical Character Recognition Using Deep Learning Techniques. https://fritz.ai/optical-character-recognition-using-deep-learning-techniques/

Geneta, P., Atienza, R., Panopio, J. D., & Pesigan, K. W. (2019). Development of Public Employment Service Office Job Portal for Municipality of Rosario, Batangas. *Academia.edu*, *3*(2456-9992). https://www.ijarp.org/published-research-papers/june2019/Development-Of-Public-Employment-Service-Office-Job-Portal-For-Municipality-Of-Rosario-Batangas.pdf

Gupit, J. J., Analyn Bajao, & Torrion, V. B. (2025). Job seekers' awareness towards the public employment services of the local government unit of Tangub City. *International Journal of Multidisciplinary Applied Business and Education Research*, *6*(6), 2725–2735. https://doi.org/10.11594/ijmaber.06.06.05

International Labour Organization. (2025, May 6). *Philippines launches 10-year labour market plan for decent work*. International Labour Organization. https://www.ilo.org/resource/news/philippines-launches-10-year-labour-market-plan-decent-work

International Telecommunication Union. (2024, November 27). *Global internet use continues to rise but disparities remain, especially in low-income regions*. International Telecommunication Union. https://www.itu.int/en/mediacentre/Pages/PR-2024-11-27-facts-and-figures.aspx

Jou, Y.-T., Mariñas, K. A., Saflor, C. S., Baleña, A., Gutierrez, C. J., Dela Fuente, G., Manzano, H. M., Tanglao, M. S., Verde, N. A., Alvarado, P., & Young, M. N. (2024). Investigating various factors influencing the accessibility of digital government with eGov PH mobile application. *Sustainability*, *16*(3), 992. https://doi.org/10.3390/su16030992

Jošt, G., & Taneski, V. (2025). State-of-the-art cross-platform mobile application development frameworks: A comparative study of market and developer trends. Informatics, 12(2), 45. https://doi.org/10.3390/informatics12020045

Marfil, R., & Ong, K. A. (2023). Public employment service system: Job search and application for General Mariano Alvarez Cavite. *Eulogio "Amang" Rodriguez Institute of Science and Technology*, *XXIII*(34). EARIST Research Journal. https://earist.edu.ph/wp-content/uploads/ERJ-2023-Volume-XXIII-No.-34-July-December.pdf#page=140

Modak, S., Shinde, P., Tiwari, A., & Nalamwar, S. (2024). *A review of resume analysis and job description matching using machine learning.* International Journal on Recent and Innovation Trends in Computing and Communication, 12(2), 247-250. Retrieved from https://ijritcc.org/index.php/ijritcc/article/view/10561.

Moumane, K., Idri, A., & Abran, A. (2024). Usability evaluation of mobile applications using ISO 25010. Software Quality Journal.

Parinsi, M. T., Palilingan, V. R., Kembuan, O., & Ratumbuisang, K. F. (2020). Job seeker information system using online web based and android mobile phones. *IOP Conference Series: Materials Science and Engineering*, *830*(022093), 022093. https://doi.org/10.1088/1757-899x/830/2/022093

Philippine Statistics Authority. (2025, November 6). *Participation in the labor force in September 2025 decreased to 51.56 million Filipinos aged 15 years and over*. Philippine Statistics Authority. https://psa.gov.ph/content/participation-labor-force-september-2025-decreased-5156-million-filipinos-aged-15-years-and

The Report Cube. (2025). Philippines smartphone market report. https://www.thereportcubes.com/report-store/philippines-smartphone-market

Venkatesh, V., Morris, M. G., Davis, G. B., & Davis, F. D. (2003). User acceptance of information technology: Toward a unified view. *MIS quarterly*, *27*(3), pp.425-478.